BLOG-DOC-08

Complete PostgreSQL Data Model, Schema, Relationships, Constraints, RLS & Seed Specification

Wave-2 owner-authorized persistence reconciliation

Revision	v1.0-RC2
Date	11 September 2026
Status	OWNER-AUTHORIZED WAVE-2 RECONCILIATION
Supersedes	v1.0-RC1 only where this reconciliation explicitly says so
Authority rule	This revision resolves existing contradictions only; it creates no new product scope.

Normative reconciliation decisions

CB-W2-01 post_autosaves

DB-TBL-021 post_autosaves adds editor_instance_id text NOT NULL. The per-editor uniqueness rule becomes UNIQUE(site_id, post_id, editor_admin_id, editor_instance_id). A length CHECK enforces 1-128 characters. Existing rows are forward-migrated with deterministic legacy identifiers. RLS and site ownership are unchanged.

CB-W2-02 structured_data_inputs

posts.structured_data_inputs remains JSONB with an object-type invariant. RC2 explicitly aligns its semantic mapping to DOC-07 RC2: object keyed by generated object type, with validated scalar/scalar-array leaf values. No JSONB column or table change is required for this blocker.

CB-W2-04 Homepage lead Post

DB-TBL-029 site_settings adds nullable homepage_lead_post_id UUID. The field uses a composite FOREIGN KEY (site_id, homepage_lead_post_id) -> posts(site_id, post_id), preserving same-site ownership. Application eligibility requires the selected lead Post to be currently PUBLISHED before assignment. Permanent Post deletion clears any matching Homepage lead reference before deleting the Post.

CB-W2-05 structured Site Settings

Existing JSONB fields contact_info, social_profiles, seo_defaults and analytics_verification remain JSONB objects and are transported as JSON-compatible objects rather than JSON-encoded strings. No new settings table is created.

Invariant / count effect

Physical estate remains 33 tables = 20 SITE-owned FORCE-RLS tables + 13 GLOBAL/system tables. Permission seed count remains 69. The forward-only reconciliation migration is DB-MIG-017; historical migrations are not rewritten.

Incorporation rule

Supersedes only affected persistence clauses in Sections 9, 11, 12, 17, 19, 22 and the relevant appendices of RC1. All unaffected RC1 DDL/RLS/seed semantics remain incorporated unchanged.

Source inclusion note

The following pages are the complete inherited v1.0-RC1 source document. They remain normative except where a clause is explicitly superseded by the reconciliation pages above. This packaging avoids silently rewriting historical source pages while issuing the controlling v1.0-RC2 contract.

BLOG-DOC-08

Complete PostgreSQL Data Model,

Schema, Relationships, Constraints, RLS & Seed Specification

Complete persistence authority for the centralized 14-site blogging platform

33	20	69	14
canonical tables	site-owned RLS tables	fixed permission seeds	initial site seeds

CORE PERSISTENCE PRINCIPLE

One centralized PostgreSQL architecture stores both global platform records and strictly site-owned records. Site ownership is explicit in relational keys and RLS; shared tables never mean shared tenant ownership. Every database rule in this document physically implements an already-frozen product, authorization, feature, IA, isolation or content decision.

Document control	Value
Status	REVIEW CANDIDATE - not frozen until approved
Version	1.0-RC1
Date	09 September 2026
Frozen upstream baseline	BLOG-DOC-01 through BLOG-DOC-07, all v1.0-RC1
Database estate	1 centralized PostgreSQL logical architecture; configuration-driven beyond 14 sites
Schema baseline	33 canonical tables = 20 SITE-owned + 13 GLOBAL/system
RLS baseline	All 20 site-owned tables RLS-enabled; normal runtime has no BYPASSRLS
Permission seed	69 keys = 44 SITE + 25 GLOBAL
Initial site seed	SITE-01 through SITE-14 with fixed stable UUIDs and canonical domains
Change policy	After freeze, table/column/constraint/RLS/seed changes require versioned migration and traceability update.

Developer lock target

A developer must never decide ad hoc whether site_id is nullable on a site-owned record, whether same-site relationships are merely an application convention, whether an unknown permission may be inserted, whether a stale Post update may overwrite a newer version, whether a pooled connection may retain tenant context, whether a production table may be edited manually, or which seed data is required. Those persistence decisions are locked here.

Document Map

Section	Subject
1	Purpose, authority and boundary
2	Persistence doctrine and one-source-of-truth rules
3	Physical PostgreSQL conventions
4	Complete ERD and table inventory
5	Tenancy and domain registry schema
6	Admin identity, roles, permissions and assignments
7	Taxonomy and author schema
8	Media persistence schema
9	Post aggregate schema
10	Post relationships
11	Formal revisions, autosaves and URL history
12	Navigation and conditional homepage persistence
13	Site and platform settings persistence
14	Redirect persistence
15	Audit and site-deletion-request persistence
16	Same-site relational integrity
17	Row-Level Security architecture
18	Database roles, grants and global-read boundary
19	Optimistic concurrency and transactional mutation rules
20	Indexing and query-shape contract
21	Archive, delete and retention behavior
22	Migration and bootstrap order
23	System and permission seed contract
24	Initial 14-site seed contract
25	Development and isolation-test seed contract
26	Database acceptance criteria
27	Traceability and upstream correlation
28	Downstream dependencies and deliberate exclusions
29	Cross-document consistency gate and final lock
Appendix A	Canonical 33-table registry
Appendix B	69-permission seed registry
Appendix C	Permission-prerequisite seed registry
Appendix D	14-site seed registry
Appendix E	RLS policy matrix
Appendix F	One-page developer execution reference

AUTHORITY BOUNDARY

THIS DOCUMENT OWNS: physical tables/columns/types, keys and constraints, site-aware relational integrity, database RLS,

runtime DB-role boundaries, persistence-side retention behavior, migration/bootstrap ordering, exact initial system/permission/site seeds and database acceptance tests. THIS DOCUMENT DOES NOT OWN: final URL route patterns (PDF 09), exact Admin/Public flows (PDFs 10/11), public response projection (PDF 13), HTTP transport (PDF 15), search ranking/index implementation (PDF 16), object-storage/image processing mechanics (PDF 17), queue/cache job payloads (PDF 18), credential/session/MFA secrets (PDF 19), performance/RPO/RTO budgets (PDF 20), repository/ORM choice (PDF 23), or deployment orchestration (PDF 24).

1. Purpose, Authority and Boundary

1.1 Purpose

This specification converts the already-frozen product, authorization, feature, site-isolation and content contracts into one deterministic PostgreSQL persistence model. It tells the implementation team exactly which durable records exist, which records are global versus site-owned, how same-site ownership is physically enforced, how the five-state Post lifecycle is stored, how revisions differ from autosaves, how role/permission scope is persisted, how tenant RLS is applied and how a clean environment is bootstrapped reproducibly.

1.2 Authority order

Priority	Authority	Controls
1	BLOG-DOC-01	Product topology, 14-site estate, centralized PostgreSQL direction and product invariants.
2	BLOG-DOC-02	Admin identity, account states, 69 permission keys, role scopes, assignments and authorization semantics.
3	BLOG-DOC-03	161 approved capabilities and SITE/GLOBAL feature scope.
4	BLOG-DOC-04	Trusted SiteContext, site/global ownership, same-site integrity, RLS direction and isolation walls.
5	BLOG-DOC-05/06	Frozen Public and Admin page inventories; DB must support them without inventing page types/modules.
6	BLOG-DOC-07	Post logical fields, ArticleBody, five persisted states, revisions/autosave/URL history/concurrency/workflow.
7	BLOG-DOC-08 - this document	Physical persistence implementation of those decisions.
8	Later specialist authorities	May refine their own domains but must use this persistence contract or formally reopen it.

1.3 Conflict rule

CONFLICT RULE

If a physical schema choice would require a new product feature, new permission, new page/module, cross-site ownership, hidden lifecycle state or different Post semantics, implementation stops. DOC-08 does not silently change an upstream frozen decision. Conversely, later documents may not weaken non-null site ownership, same-site FKs, RLS or seed/constraint invariants without reopening this persistence authority.

2. Persistence Doctrine and One-Source-of-Truth Rules

Principle ID	Locked persistence rule
DB-PR-01	One logical PostgreSQL architecture; no per-site database/table clones.
DB-PR-02	Global and site-owned records may share infrastructure but never ownership semantics.

Principle ID	Locked persistence rule
DB-PR-03	Every site-owned primary/child table carries non-null site_id, except no deliberately global table masquerades as site-owned.
DB-PR-04	Same-site relationships use composite site_id foreign keys where both endpoints are site-owned.
DB-PR-05	site_id is server-derived and immutable in ordinary operations.
DB-PR-06	RLS is defense in depth on all canonical site-owned tables; application scoping remains mandatory.
DB-PR-07	Normal runtime DB role has no BYPASSRLS, superuser or table-owner bypass.
DB-PR-08	Missing transaction tenant setting gives no site-owned access; it never means all sites.
DB-PR-09	Global management is an explicit platform path; global authority never rewrites site ownership.
DB-PR-10	No generic Pages table, capability table, nested-role table, deny table or cross-site sharing table exists in v1.
DB-PR-11	All production schema changes are migrations; hidden manual DDL is prohibited.
DB-PR-12	Required seed data is idempotent and environment-classified.
DB-PR-13	Post durable mutations use optimistic compare-and-swap and formal revision insertion.
DB-PR-14	Autosaves are private recovery state and never public truth.
DB-PR-15	Database checks enforce structure/scope; feature/state/permission business rules remain in authoritative services where SQL alone cannot express them safely.

3. Physical PostgreSQL Conventions

3.1 Naming and identifiers

Concern	Frozen convention
Tables/columns/indexes	lowercase snake_case.
Primary IDs	uuid with gen_random_uuid() unless a stable natural key is explicitly authoritative (permission_key, setting_key).
Timestamps	timestamptz stored as UTC instants.
Local scheduling input	timestamp without time zone plus IANA timezone ID plus resolved UTC instant.
Statuses	text + CHECK constraints; no PostgreSQL ENUM types in v1 to keep controlled migrations explicit.
Human strings	text; length/business validation is service-side unless a DB safety check is listed.
Structured payloads	jsonb only where upstream structure is inherently document-like; JSON type checks are mandatory.
Generated normalization	stored generated columns such as email_normalized/name_normalized where DB uniqueness depends on normalized comparison.
Constraint names	pk_, fk_, uq_, ck_ prefixes; indexes ix_.
Site composite FK pattern	Parent exposes UQ(site_id, <id>); child carries site_id and references both columns.

3.2 Required extension and helper schema

```
CREATE EXTENSION IF NOT EXISTS pgcrypto;
```

```
CREATE SCHEMA IF NOT EXISTS app;

CREATE FUNCTION app.current_site_id() RETURNS uuid
LANGUAGE sql STABLE AS $$
  SELECT NULLIF(current_setting('app.site_id', true), '')::uuid;
$$;
```

The helper returns NULL when no tenant context exists. Site-owned RLS predicates compare site_id to app.current_site_id(); NULL therefore grants no tenant rows. The application sets app.site_id with SET LOCAL inside a transaction after trusted SiteContext resolution. The setting is never persisted across transaction boundaries.

3.3 Site-ownership immutability trigger

```
CREATE FUNCTION app.reject_site_id_change() RETURNS trigger AS $$
BEGIN
  IF NEW.site_id IS DISTINCT FROM OLD.site_id THEN
    RAISE EXCEPTION 'site_id is immutable';
  END IF;
  RETURN NEW;
END; $$ LANGUAGE plpgsql;
```

The migration applies this trigger to every site-owned table whose rows are ordinarily mutable. Junction rows are replaced rather than re-homed. RLS WITH CHECK independently prevents cross-tenant INSERT/UPDATE.

4. Complete ERD and Canonical Table Inventory

The complete model contains 33 canonical tables. Twenty are site-owned and RLS-protected; thirteen are global/system tables. The diagrams below are complementary views of one ERD rather than separate data models.

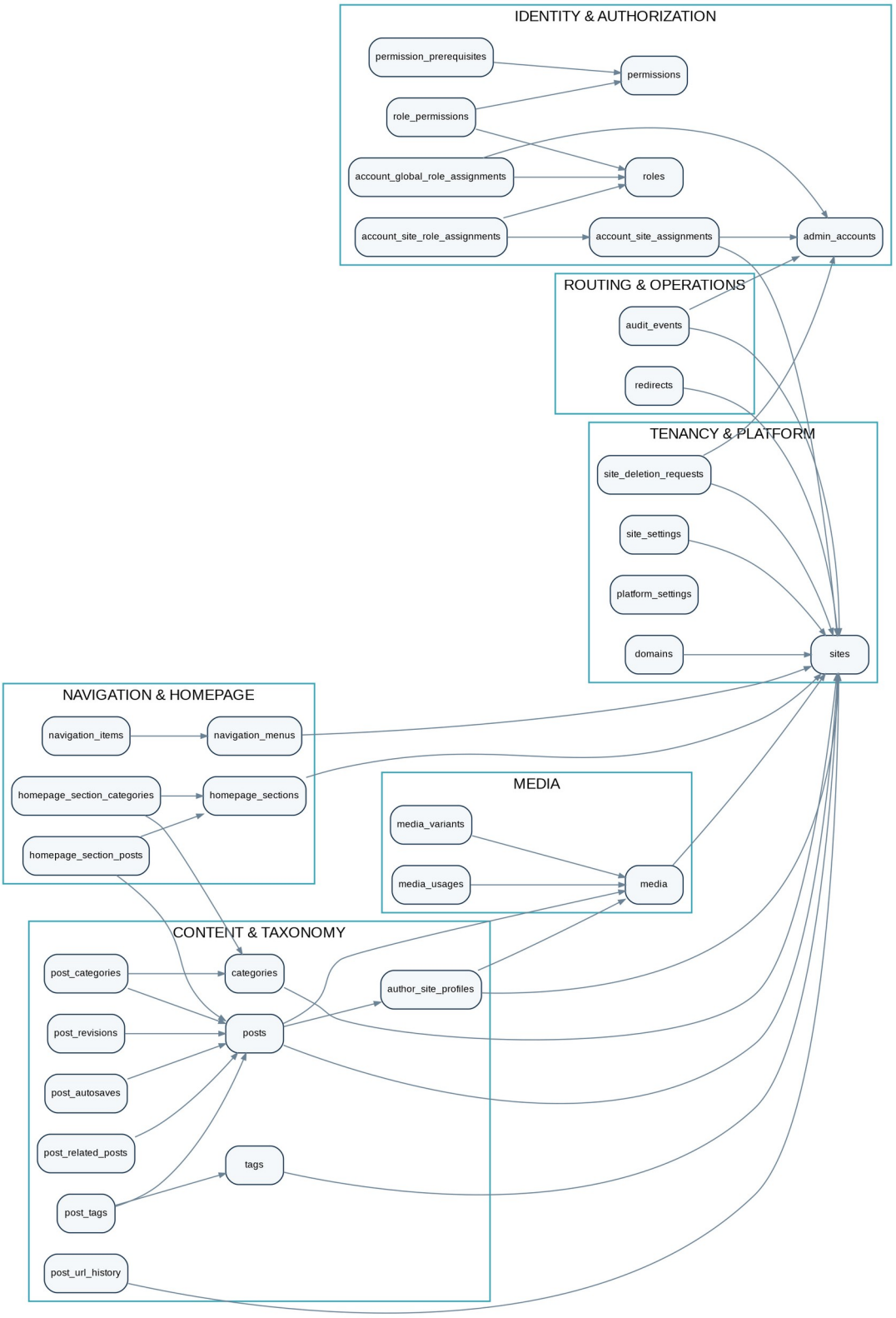


Figure 1 - Complete subsystem ERD overview (all 33 canonical tables). Detailed subsystem ERDs follow.

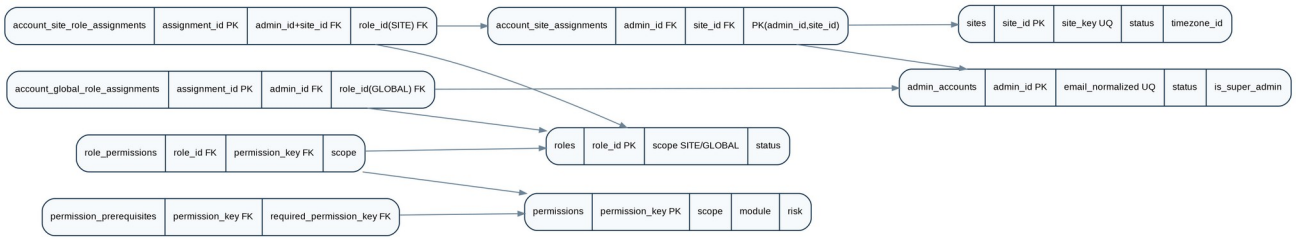


Figure 2 - Identity, site assignment, role and permission relationship ERD.

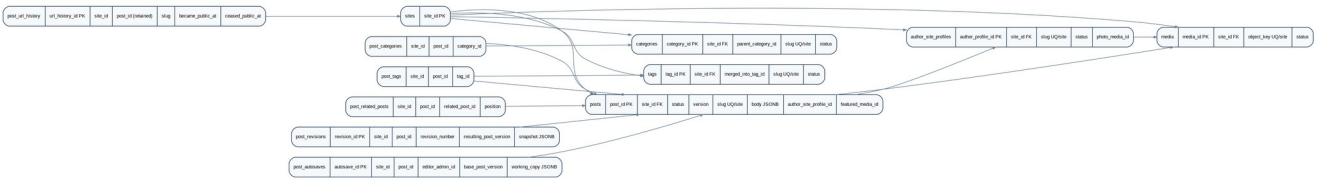


Figure 3 - Post, taxonomy, author, media, revision/autosave and URL-history ERD.

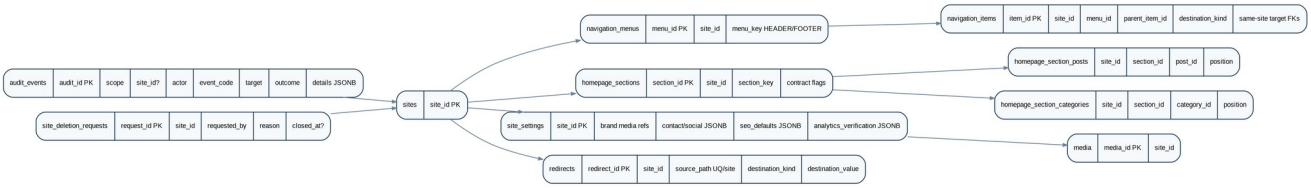


Figure 4 - Navigation, homepage, settings, redirects, audit and deletion-request ERD.

4.1 Inventory arithmetic

Class	Count	RLS	Meaning
SITE-owned	20	YES	Rows belong to exactly one site and are protected by RLS + same-site constraints.
GLOBAL/system	13	NO tenant RLS	Platform/authorization/registry/ audit records with explicit global semantics.
TOTAL	33	-	Canonical v1 persistence baseline.

5. Tenancy and Domain Registry Schema

DB-TBL-001 - sites

Property	Locked value
Scope	GLOBAL
RLS	NO
Retention/delete	Lifecycle retained; no ordinary hard delete.

Column	PostgreSQL type	NULL?	Default	Physical meaning
site_id	uuid	NO	gen_random_uuid()	Stable internal site identity.
site_key	text	NO	-	Stable human-readable key such as SITE-01.
display_name	text	NO	-	Admin/public identity label; mutable.
status	text	NO	'DISABLED'	ACTIVE | DISABLED | ARCHIVED | DELETION_REQUESTED.
timezone_id	text	NO	-	Required IANA timezone identifier supplied by approved site bootstrap/configuration; no portfolio-wide timezone is assumed.
locale_code	text	YES	NULL	Configured locale when present.
default_language_code	text	YES	NULL	Configured default language when present.
disabled_at	timestamptz	YES	NULL	Current disable transition timestamp.
archived_at	timestamptz	YES	NULL	Current archive transition timestamp.
deletion_requested_at	timestamptz	YES	NULL	Current deletion-request transition timestamp.
created_at	timestamptz	NO	now()	Registry creation time.
updated_at	timestamptz	NO	now()	Latest registry mutation time.

Constraints

- PK(site_id)
- UQ(site_key)
- CK site_key format ^SITE-[0-9]{2,}$
- CK status in four frozen lifecycle values
- CK locale/language are nonblank when present

Indexes

- uq_sites__site_key
- ix_sites__status

> **IMPLEMENTATION RULE**
>
> Initial 14-site seed requires an explicit valid IANA timezone value for every site; DOC-01 through DOC-07 do not freeze those concrete timezone values. New sites default DISABLED until global lifecycle rules enable them.

DB-TBL-002 - domains

Property	Locked value
Scope	GLOBAL
RLS	NO

Property	Locked value
Retention/delete	Rows retained while mapped; aliases may be removed under global domain management.

Column	PostgreSQL type	NULL?	Default	Physical meaning
domain_id	uuid	NO	gen_random_uuid()	Stable mapping identity.
site_id	uuid	NO	-	Owning site registry identity.
hostname	text	NO	-	Stored hostname input.
hostname_normalized	text	NO	generated lower(btrim(hostname))	Canonical comparison key; PDF 09 owns full normalization.
is_canonical	boolean	NO	false	True for the site canonical host.
is_active	boolean	NO	true	Whether resolver may consider mapping.
created_at	timestamptz	NO	now()	Mapping creation time.
updated_at	timestamptz	NO	now()	Latest mapping mutation time.

Constraints

- PK(domain_id)
- FK site_id -> sites(site_id) ON DELETE CASCADE
- UQ(hostname_normalized) across estate
- Partial UQ(site_id) WHERE is_canonical AND is_active
- CK hostname has no scheme/path whitespace

Indexes

- uq_domains__hostname_normalized
- uq_domains__one_active_canonical_per_site
- ix_domains__site_id_active

> **IMPLEMENTATION RULE**
>
> A site may exist while disabled before a canonical domain is configured. Enabling is validated by later lifecycle/routing authority.

6. Admin Identity, Roles, Permissions and Assignments

DB-TBL-003 - admin_accounts

Property	Locked value
Scope	GLOBAL
RLS	NO
Retention/delete	Disable instead of routine hard delete; historical identity retained.

Column	PostgreSQL type	NULL?	Default	Physical meaning
admin_id	uuid	NO	gen_random_uuid()	Stable Admin identity.
email	text	NO	-	Human login identifier.
email_normalized	text	NO	generated lower(btrim(email))	Case-insensitive uniqueness key.
display_name	text	NO	-	Mutable display name.
status	text	NO	'INVITED'	INVITED | ACTIVE | DISABLED.
is_super_admin	boolean	NO	false	Designated owner-level principal flag.
created_at	timestamptz	NO	now()	Identity creation time.
updated_at	timestamptz	NO	now()	Latest identity/profile mutation.
disabled_at	timestamptz	YES	NULL	Disable timestamp when applicable.

Constraints

- PK(admin_id)
- UQ(email_normalized)
- Partial UQ(is_super_admin) WHERE is_super_admin=true
- CK status in frozen account states
- CK disabled_at present iff status=DISABLED

Indexes

- uq_admin_accounts__email_normalized
- uq_admin_accounts__single_super_admin
- ix_admin_accounts__status

> **IMPLEMENTATION RULE**
>
> Credential hashes, MFA secrets, reset/invitation tokens and sessions are intentionally not defined here; PDF 19 owns those security persistence decisions.

DB-TBL-004 - permissions

Property	Locked value
Scope	GLOBAL
RLS	NO
Retention/delete	Migration/seed-owned; runtime Admin is read-only.

Column	PostgreSQL type	NULL?	Default	Physical meaning
permission_key	text	NO	-	Stable machine key; primary identifier.
scope	text	NO	-	SITE | GLOBAL.
module_key	text	NO	-	Human/module grouping from DOC-02.
risk	text	NO	-	Standard | Sensitive | Destructive.

Column	PostgreSQL type	NULL?	Default	Physical meaning
label	text	NO	-	Human-readable label/help name.
description	text	NO	-	Canonical help description.
seed_version	integer	NO	1	Version of registry row.
created_at	timestamptz	NO	now()	Seed insertion timestamp.

Constraints

- PK(permission_key)
- UQ(permission_key,scope)
- CK scope SITE/GLOBAL
- CK risk STANDARD/SENSITIVE/DESTRUCTIVE

Indexes

- ix_permissions__scope_module

> **IMPLEMENTATION RULE**
>
> Exactly 69 rows are seeded: 44 SITE + 25 GLOBAL. No wildcard/deny keys exist.

DB-TBL-005 - permission_prerequisites

Property	Locked value
Scope	GLOBAL
RLS	NO
Retention/delete	Migration/seed-owned; delete only through controlled permission-registry migration.

Column	PostgreSQL type	NULL?	Default	Physical meaning
permission_key	text	NO	-	Permission requiring another exact key.
required_permission_key	text	NO	-	Required prerequisite key.
created_at	timestamptz	NO	now()	Seed timestamp.

Constraints

- PK(permission_key,required_permission_key)
- FK both keys -> permissions(permission_key) ON DELETE RESTRICT
- CK permission_key <> required_permission_key

Indexes

- ix_permission_prerequisites__required

> **IMPLEMENTATION RULE**
>
> Stores DOC-02 role-composition prerequisites only; capability-specific composite authorization remains DOC-02 authority and is not redefined as database data.

DB-TBL-006 - roles

Property	Locked value
Scope	GLOBAL
RLS	NO
Retention/delete	Retired, not routine hard-deleted after use.

Column	PostgreSQL type	NULL?	Default	Physical meaning
role_id	uuid	NO	gen_random_uuid()	Stable reusable role identity.
scope	text	NO	-	SITE | GLOBAL; immutable once role is used.
name	text	NO	-	Human role name; not security identity.
name_normalized	text	NO	generated lower(btrim(name))	Uniqueness key within scope.
description	text	YES	NULL	Operational description.
status	text	NO	'ACTIVE'	ACTIVE | RETIRED.
created_by_admin_id	uuid	NO	-	Creating Admin.
updated_by_admin_id	uuid	YES	NULL	Latest mutating Admin.
created_at	timestamptz	NO	now()	Creation time.
updated_at	timestamptz	NO	now()	Latest mutation.
retired_at	timestamptz	YES	NULL	Retirement time.

Constraints

- PK(role_id)
- UQ(role_id,scope)
- UQ(scope,name_normalized)
- FK actor IDs -> admin_accounts
- CK scope SITE/GLOBAL
- CK status ACTIVE/RETIRED
- CK retired_at iff RETIRED

Indexes

- uq_roles__scope_name
- ix_roles__scope_status

> **IMPLEMENTATION RULE**
>
> Database trigger blocks scope mutation after first permission or assignment, and blocks retirement while active assignments remain.

DB-TBL-007 - role_permissions

Property	Locked value
Scope	GLOBAL
RLS	NO
Retention/delete	Rows change only through authorized role edits; removed rows are audited externally.

Column	PostgreSQL type	NULL?	Default	Physical meaning
role_id	uuid	NO	-	Role receiving grant.
permission_key	text	NO	-	Exact permission key.
scope	text	NO	-	Duplicated scope solely for composite FK enforcement.
created_at	timestamptz	NO	now()	Grant insertion time.

Constraints

- PK(role_id,permission_key)
- FK(role_id,scope) -> roles(role_id,scope)
- FK(permission_key,scope) -> permissions(permission_key,scope)
- CK scope SITE/GLOBAL

Indexes

- ix_role_permissions__permission_key

IMPLEMENTATION RULE

Scope mismatch is impossible at the FK layer; prerequisite consistency is validated transactionally before commit.

DB-TBL-008 - account_site_assignments

Property	Locked value
Scope	GLOBAL
RLS	NO
Retention/delete	Hard delete represents assignment removal; audit preserves history.

Column	PostgreSQL type	NULL?	Default	Physical meaning
admin_id	uuid	NO	-	Assigned Admin.
site_id	uuid	NO	-	Site membership.
assigned_by_admin_id	uuid	NO	-	Authorized assigning Admin.
assigned_at	timestamptz	NO	now()	Assignment time.

Constraints

- PK(admin_id,site_id)
- FK admin_id -> admin_accounts ON DELETE RESTRICT
- FK site_id -> sites ON DELETE RESTRICT
- FK assigned_by_admin_id -> admin_accounts ON DELETE RESTRICT

Indexes

- ix_account_site_assignments__site_id

IMPLEMENTATION RULE

Membership only; it grants no action permission. Removing a row cascades associated SITE role assignments.

DB-TBL-009 - account_site_role_assignments

Property	Locked value
Scope	GLOBAL
RLS	NO
Retention/delete	Hard delete represents role unassignment; audit preserves history.

Column	PostgreSQL type	NULL?	Default	Physical meaning
assignment_id	uuid	NO	gen_random_uuid()	Assignment identity.
admin_id	uuid	NO	-	Assigned Admin.
site_id	uuid	NO	-	Exact site scope.
role_id	uuid	NO	-	SITE role.
scope	text	NO	'SITE'	Constant discriminator for composite FK.
assigned_by_admin_id	uuid	NO	-	Authorized assigning Admin.
assigned_at	timestamptz	NO	now()	Assignment time.

Constraints

- PK(assignment_id)
- UQ(admin_id,site_id,role_id)
- FK(admin_id,site_id) -> account_site_assignments ON DELETE CASCADE
- FK(role_id,scope) -> roles(role_id,scope)

- CK scope=SITE

Indexes

- ix_account_site_role_assignments__admin_site
- ix_account_site_role_assignments__role

> **IMPLEMENTATION RULE**
>
> Site assignment is structurally required before a SITE role can exist.

DB-TBL-010 - account_global_role_assignments

Property	Locked value
Scope	GLOBAL
RLS	NO
Retention/delete	Hard delete represents role unassignment; audit preserves history.

Column	PostgreSQL type	NULL?	Default	Physical meaning
assignment_id	uuid	NO	gen_random_uuid()	Assignment identity.
admin_id	uuid	NO	-	Assigned Admin.
role_id	uuid	NO	-	GLOBAL role.
scope	text	NO	'GLOBAL'	Constant discriminator for composite FK.
assigned_by_admin_id	uuid	NO	-	Authorized assigning Admin.
assigned_at	timestamptz	NO	now()	Assignment time.

Constraints

- PK(assignment_id)
- UQ(admin_id,role_id)
- FK admin_id -> admin_accounts
- FK(role_id,scope) -> roles(role_id,scope)
- CK scope=GLOBAL

Indexes

- ix_account_global_role_assignments__admin
- ix_account_global_role_assignments__role

> **IMPLEMENTATION RULE**
>
> GLOBAL roles do not create site membership and do not bypass SiteContext for site-owned objects.

7. Taxonomy and Author Schema

DB-TBL-011 - categories

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Soft-retired in normal product flow; no routine hard purge.

Column	PostgreSQL type	NULL?	Default	Physical meaning
category_id	uuid	NO	gen_random_uuid()	Stable category identity.
site_id	uuid	NO	-	Owning site.
parent_category_id	uuid	YES	NULL	Optional same-site parent.
name	text	NO	-	Category name.
name_normalized	text	NO	generated lower(btrim(name))	Site uniqueness key.
slug	text	NO	-	Site-scoped slug; PDF 09 owns normalization.
description	text	YES	NULL	Editorial/category description.
seo_title	text	YES	NULL	Approved category SEO input.
meta_description	text	YES	NULL	Approved category SEO input.
canonical_override	text	YES	NULL	Optional approved override input.
index_policy	text	NO	'INHERIT'	INHERIT | INDEX | NOINDEX.
status	text	NO	'ACTIVE'	ACTIVE | RETIRED.
retired_at	timestamptz	YES	NULL	Retirement time.
created_at	timestamptz	NO	now()	Creation time.
updated_at	timestamptz	NO	now()	Latest mutation.

Constraints

- PK(category_id)
- UQ(site_id,category_id)
- FK site_id -> sites
- Composite self-FK(site_id,parent_category_id) -> categories
- UQ(site_id,name_normalized)
- UQ(site_id,slug)
- CK parent <> self
- CK status ACTIVE/RETIRED
- CK retired_at iff RETIRED
- CK index_policy allowed

Indexes

- uq_categories__site_slug
- uq_categories__site_name
- ix_categories__site_parent
- ix_categories__site_status

> **IMPLEMENTATION RULE**
>
> Parent/child and Post assignment are same-site by composite FK. Retired categories remain addressable for history but are not eligible for new assignment.

DB-TBL-012 - tags

Property	Locked value
Scope	SITE

Property	Locked value
RLS	YES
Retention/delete	Soft-retired; tag merge retires the source after relationship rewrite.

Column	PostgreSQL type	NULL?	Default	Physical meaning
tag_id	uuid	NO	gen_random_uuid()	Stable tag identity.
site_id	uuid	NO	-	Owning site.
name	text	NO	-	Tag name.
name_normalized	text	NO	generated lower(btrim(name))	Site uniqueness key.
slug	text	NO	-	Site-scoped slug.
description	text	YES	NULL	Optional public/editorial description.
status	text	NO	'ACTIVE'	ACTIVE | RETIRED.
merged_into_tag_id	uuid	YES	NULL	Same-site target when retired by merge.
retired_at	timestamptz	YES	NULL	Retirement time.
created_at	timestamptz	NO	now()	Creation time.
updated_at	timestamptz	NO	now()	Latest mutation.

Constraints

- PK(tag_id)
- UQ(site_id,tag_id)
- FK site_id -> sites
- Composite FK(site_id,merged_into_tag_id) -> tags
- UQ(site_id,name_normalized)
- UQ(site_id,slug)
- CK merged target <> self
- CK status ACTIVE/RETIRED
- CK merged_into_tag_id implies RETIRED

Indexes

- uq_tags__site_slug
- uq_tags__site_name
- ix_tags__site_status

> **IMPLEMENTATION RULE**
>
> Merge is a transaction: move same-site post associations, de-duplicate via PK, mark source RETIRED and set merged_into_tag_id; no posts are deleted.

DB-TBL-015 - author_site_profiles

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Soft-retired; historical Post attribution remains.

Column	PostgreSQL type	NULL?	Default	Physical meaning
author_profile_id	uuid	NO	gen_random_uuid()	Stable site-local author profile identity.
site_id	uuid	NO	-	Owning site.
display_name	text	NO	-	Public author name.
slug	text	NO	-	Site-scoped public author slug.

Column	PostgreSQL type	NULL?	Default	Physical meaning
photo_media_id	uuid	YES	NULL	Same-site author photo.
bio	text	YES	NULL	Public bio.
designation	text	YES	NULL	Public designation.
expertise	text[]	NO	'{}'	Approved expertise labels.
social_profiles	jsonb	NO	'{}'::jsonb	Validated social-profile map.
seo_title	text	YES	NULL	Author-page SEO title input.
meta_description	text	YES	NULL	Author-page meta description input.
canonical_override	text	YES	NULL	Optional canonical override input.
index_policy	text	NO	'INHERIT'	INHERIT | INDEX | NOINDEX.
status	text	NO	'ACTIVE'	ACTIVE | RETIRED.
retired_at	timestamptz	YES	NULL	Retirement time.
created_at	timestamptz	NO	now()	Creation time.
updated_at	timestamptz	NO	now()	Latest mutation.

Constraints

- PK(author_profile_id)
- UQ(site_id,author_profile_id)
- FK site_id -> sites
- Composite FK(site_id,photo_media_id) -> media
- UQ(site_id,slug)
- CK JSON social_profiles object
- CK status ACTIVE/RETIRED
- CK index_policy allowed

Indexes

- uq_author_profiles__site_slug
- ix_author_profiles__site_status
- ix_author_profiles__site_name

IMPLEMENTATION RULE

There is no global cross-site Author table in v1; each profile is site-owned, matching isolation and the absence of a global Authors module.

8. Media Persistence Schema

DB-TBL-013 - media

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Soft-deleted/tombstoned in normal product flow; physical object purge is later media/security policy.

Column	PostgreSQL type	NULL?	Default	Physical meaning
media_id	uuid	NO	gen_random_uuid()	Stable media identity.
site_id	uuid	NO	-	Owning site.
status	text	NO	'ACTIVE'	ACTIVE | DELETED.
original_filename	text	NO	-	Original upload filename.
mime_type	text	NO	-	Validated MIME type.
byte_size	bigint	NO	-	Original byte size > 0.
width_px	integer	YES	NULL	Pixel width when applicable.
height_px	integer	YES	NULL	Pixel height when applicable.
checksum_sha256	char(64)	NO	-	Hex SHA-256 checksum.
object_key	text	NO	-	Site-namespaced object storage key.
alt_text	text	YES	NULL	Default alt text.
title	text	YES	NULL	Editorial media title.
caption	text	YES	NULL	Default caption.
credit_text	text	YES	NULL	Credit/source display text.
source_url	text	YES	NULL	Optional source reference URL.
created_by_admin_id	uuid	NO	-	Uploader.
updated_by_admin_id	uuid	YES	NULL	Latest metadata editor.
deleted_by_admin_id	uuid	YES	NULL	Delete actor.
created_at	timestamptz	NO	now()	Upload metadata creation.
updated_at	timestamptz	NO	now()	Latest metadata mutation.
deleted_at	timestamptz	YES	NULL	Soft deletion time.

Constraints

- PK(media_id)
- UQ(site_id,media_id)
- FK site_id -> sites
- UQ(site_id,object_key)
- FK actor IDs -> admin_accounts
- CK status ACTIVE/DELETED
- CK byte_size>0
- CK dimensions positive when present
- CK deleted_at/by required for DELETED

Indexes

- uq_media__site_object_key
- ix_media__site_status_created
- ix_media__site_checksum

> **IMPLEMENTATION RULE**
>
> Object key must be namespaced by site. New references may use ACTIVE media only. Existing historical revision snapshots may

reference tombstoned media until current validation resolves replacement/restore behavior.

DB-TBL-014 - media_variants

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Cascade with media tombstone/purge policy; variants are derived.

Column	PostgreSQL type	NULL?	Default	Physical meaning
variant_id	uuid	NO	gen_random_uuid()	Variant identity.
site_id	uuid	NO	-	Inherited site.
media_id	uuid	NO	-	Owning media.
variant_key	text	NO	-	Stable variant purpose/key.
mime_type	text	NO	-	Variant MIME type.
byte_size	bigint	NO	-	Variant bytes > 0.
width_px	integer	YES	NULL	Variant width.
height_px	integer	YES	NULL	Variant height.
object_key	text	NO	-	Site-namespaced storage key.
created_at	timestamptz	NO	now()	Generation time.

Constraints

- PK(variant_id)
- UQ(site_id,variant_id)
- Composite FK(site_id,media_id) -> media
- UQ(site_id,media_id,variant_key)
- UQ(site_id,object_key)
- CK byte_size>0

Indexes

- uq_media_variants__media_key
- uq_media_variants__site_object_key

IMPLEMENTATION RULE

Exact variant catalog and image-processing rules remain PDF 17; this table is the persistent derived-record contract.

DB-TBL-023 - media_usages

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Derived usage rows removed/rebuilt with owning references; media deletion checks this table.

Column	PostgreSQL type	NULL?	Default	Physical meaning
usage_id	uuid	NO	gen_random_uuid()	Usage identity.
site_id	uuid	NO	-	Tenant dimension.
media_id	uuid	NO	-	Referenced media.

Column	PostgreSQL type	NULL?	Default	Physical meaning
usage_kind	text	NO	-	POST_FEATURED | POST_BODY | POST_OG | AUTHOR_PHOTO | SITE_LOGO | SITE_FAVICON | SITE_DEFAULT_SOCIAL.
post_id	uuid	YES	NULL	Post target when usage_kind is Post-based.
author_profile_id	uuid	YES	NULL	Author target when applicable.
setting_slot	text	YES	NULL	Site-setting slot for site asset references.
block_id	text	YES	NULL	ArticleBody block ID for POST_BODY usage.
created_at	timestamptz	NO	now()	Derived usage creation time.

Constraints

- PK(usage_id)
- UQ(site_id,usage_id)
- Composite FK(site_id,media_id) -> media
- Composite FK(site_id,post_id) -> posts ON DELETE CASCADE when non-null
- Composite FK(site_id,author_profile_id) -> author_site_profiles when non-null
- CK usage_kind allowed
- CK target columns match usage_kind

Indexes

- ix_media_usages__site_media
- ix_media_usages__site_post
- ix_media_usages__site_author

IMPLEMENTATION RULE

This is a derived usage index, not the referential source of truth for direct fields. Body-media references are extracted from validated ArticleBody during commit.

9. Post Aggregate Schema

DB-TBL-016 - posts

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	TRASH is soft state; authorized Permanent Delete hard-deletes the Post after gates. URL history may outlive the Post.

Column	PostgreSQL type	NULL?	Default	Physical meaning
post_id	uuid	NO	gen_random_uuid()	PST-FLD-001 stable identity.
site_id	uuid	NO	-	PST-FLD-002 immutable owner from SiteContext.
content_schema_version	smallint	NO	1	PST-FLD-003 positive body schema version.
status	text	NO	'DRAFT'	PST-FLD-004 frozen five-state lifecycle.
version	bigint	NO	1	PST-FLD-005 monotonic optimistic token.
title	text	NO	''	PST-FLD-006; may be blank only in early Draft.
excerpt	text	YES	NULL	PST-FLD-007.
body	jsonb	NO	'{"blocks":[]}'::jsonb	PST-FLD-008 structured ArticleBody.
author_site_profile_id	uuid	YES	NULL	PST-FLD-009 same-site; required before schedule/publish.
featured_media_id	uuid	YES	NULL	PST-FLD-012 same-site featured media.
slug	text	YES	NULL	PST-FLD-014; required before public eligibility.
seo_title	text	YES	NULL	PST-FLD-015.
meta_description	text	YES	NULL	PST-FLD-016.
canonical_override	text	YES	NULL	PST-FLD-017.
index_policy	text	NO	'INHERIT'	PST-FLD-018.
link_robots_policy	text	NO	'INHERIT'	PST-FLD-019.
og_title	text	YES	NULL	PST-FLD-020.
og_description	text	YES	NULL	PST-FLD-021.
og_image_media_id	uuid	YES	NULL	PST-FLD-022 same-site media.
structured_data_inputs	jsonb	NO	'{}'::jsonb	PST-FLD-023 validated source object; no raw JSON-LD.
breadcrumb_label	text	YES	NULL	PST-FLD-024.
scheduled_for_utc	timestamptz	YES	NULL	PST-FLD-025.
schedule_local_input	timestamp without time zone	YES	NULL	PST-FLD-026 retained wall-clock input.
schedule_timezone_id	text	YES	NULL	PST-FLD-027 IANA timezone used at scheduling.
first_published_at	timestamptz	YES	NULL	PST-FLD-028 immutable after first publish.
published_at	timestamptz	YES	NULL	PST-FLD-029 latest transition into PUBLISHED.
archived_at	timestamptz	YES	NULL	PST-FLD-030 current archive timestamp.

Column	PostgreSQL type	NULL?	Default	Physical meaning
trashed_at	timestamptz	YES	NULL	PST-FLD-031 current trash timestamp.
created_at	timestamptz	NO	now()	PST-FLD-032.
created_by_admin_id	uuid	NO	-	PST-FLD-033.
updated_at	timestamptz	NO	now()	PST-FLD-034.
updated_by_admin_id	uuid	YES	NULL	PST-FLD-035; NULL means explicit SYSTEM transition.

Constraints

- PK(post_id)
- UQ(site_id,post_id)
- FK site_id -> sites
- Composite FK(site_id,author_site_profile_id) -> author_site_profiles
- Composite FK site/media for featured_media_id and og_image_media_id
- FK actor IDs -> admin_accounts
- Partial UQ(site_id,slug) WHERE slug IS NOT NULL
- CK status in DRAFT/SCHEDULED/PUBLISHED/ARCHIVED/TRASH
- CK version>0 and content_schema_version>0
- CK body is object with blocks array
- CK structured_data_inputs is object
- CK SCHEDULED iff all three schedule fields non-null; non-SCHEDULED requires all three null
- CK archived_at present iff ARCHIVED
- CK trashed_at present iff TRASH
- CK index/link policy allowed

Indexes

- uq_posts__site_slug
- ix_posts__site_status_published
- ix_posts__site_status_scheduled
- ix_posts__site_author
- ix_posts__site_updated
- ix_posts__site_created

IMPLEMENTATION RULE

Category/tag/related relations are normalized into junction tables. Every durable mutation is compare-and-swap on version; the service performs UPDATE ... WHERE site_id=? AND post_id=? AND version=expected_version.

10. Post Relationship Tables

DB-TBL-017 - post_categories

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Cascade with Post permanent delete; category retirement does not erase historical association automatically.

Column	PostgreSQL type	NULL?	Default	Physical meaning
site_id	uuid	NO	-	Shared tenant identity.
post_id	uuid	NO	-	Owning Post.
category_id	uuid	NO	-	Assigned category.
created_at	timestamptz	NO	now()	Association creation time.

Constraints

- PK(site_id,post_id,category_id)
- Composite FK(site_id,post_id) -> posts ON DELETE CASCADE
- Composite FK(site_id,category_id) -> categories ON DELETE RESTRICT

Indexes

- ix_post_categories__site_category

> **IMPLEMENTATION RULE**
>
> Composite FKs make cross-site assignment impossible.

DB-TBL-018 - post_tags

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Cascade with Post permanent delete; tag merge rewrites rows transactionally.

Column	PostgreSQL type	NULL?	Default	Physical meaning
site_id	uuid	NO	-	Shared tenant identity.
post_id	uuid	NO	-	Owning Post.
tag_id	uuid	NO	-	Assigned tag.
created_at	timestamptz	NO	now()	Association creation time.

Constraints

- PK(site_id,post_id,tag_id)
- Composite FK(site_id,post_id) -> posts ON DELETE CASCADE
- Composite FK(site_id,tag_id) -> tags ON DELETE RESTRICT

Indexes

- ix_post_tags__site_tag

> **IMPLEMENTATION RULE**
>
> Tag merge inserts target association ON CONFLICT DO NOTHING, removes source association, then retires source tag in one transaction.

DB-TBL-019 - post_related_posts

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Cascade with source Post permanent delete; related target delete is RESTRICT until relation is removed.

Column	PostgreSQL type	NULL?	Default	Physical meaning
site_id	uuid	NO	-	Shared tenant identity.
post_id	uuid	NO	-	Source Post.
related_post_id	uuid	NO	-	Same-site related Post.
position	integer	NO	-	Stable manual ordering starting at 0.
created_at	timestamptz	NO	now()	Selection time.

Constraints

- PK(site_id,post_id,related_post_id)
- UQ(site_id,post_id,position)
- Composite FK source and related -> posts
- CK post_id <> related_post_id
- CK position>=0

Indexes

- ix_post_related_posts__site_related

> **IMPLEMENTATION RULE**
>
> Public eligibility of related targets is evaluated at render time; the DB enforces same site and no self-reference.

11. Formal Revisions, Autosaves and URL History

DB-TBL-020 - post_revisions

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Immutable while Post exists; cascades on authorized Permanent Delete.

Column	PostgreSQL type	NULL?	Default	Physical meaning
revision_id	uuid	NO	gen_random_uuid()	Stable revision identity.
site_id	uuid	NO	-	Same site as Post.
post_id	uuid	NO	-	Owning Post.
revision_number	bigint	NO	-	Monotonic per Post.
source_action	text	NO	-	Create/save/schedule/publish/ update/archive/trash/restore/ revision-restore/bulk/system action.
actor_type	text	NO	-	ADMIN | SYSTEM.
actor_admin_id	uuid	YES	NULL	Admin actor when actor_type=ADMIN.
resulting_post_version	bigint	NO	-	Post version produced by revision.
snapshot	jsonb	NO	-	Immutable full committed snapshot plus workflow metadata.
created_at	timestamptz	NO	now()	Commit time.

Constraints

- PK(revision_id)
- UQ(site_id,revision_id)
- UQ(site_id,post_id,revision_number)
- UQ(site_id,post_id,resulting_post_version)
- Composite FK(site_id,post_id) -> posts ON DELETE CASCADE
- FK actor_admin_id -> admin_accounts
- CK revision_number/resulting_post_version>0
- CK actor_type ADMIN/SYSTEM and actor_admin_id presence matches
- CK snapshot is object

Indexes

- ix_post_revisions__site_post_created
- ix_post_revisions__site_post_revision

> **IMPLEMENTATION RULE**
>
> Application runtime receives INSERT/SELECT only; UPDATE/DELETE are prohibited except cascade during Permanent Delete/migration. Revision Restore creates a new row; it never rewrites an old revision.

DB-TBL-021 - post_autosaves

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Ephemeral latest working copy per Post/editor; removed after successful durable commit or Post delete.

Column	PostgreSQL type	NULL?	Default	Physical meaning
autosave_id	uuid	NO	gen_random_uuid()	Autosave identity.
site_id	uuid	NO	-	Same site as Post.
post_id	uuid	NO	-	Owning Post.
editor_admin_id	uuid	NO	-	Admin whose private working copy this is.
base_post_version	bigint	NO	-	Version against which working copy began.
content_schema_version	smallint	NO	-	Article schema version.
working_copy	jsonb	NO	-	Private recoverable editorial/relationship/SEO working copy; no lifecycle transition.
saved_at	timestamptz	NO	now()	Latest autosave time.

Constraints

- PK(autosave_id)
- UQ(site_id,post_id,editor_admin_id)
- Composite FK(site_id,post_id) -> posts ON DELETE CASCADE
- FK editor_admin_id -> admin_accounts
- CK base_post_version>0
- CK working_copy object

Indexes

- uq_post_autosaves__post_editor
- ix_post_autosaves__editor_saved

> **IMPLEMENTATION RULE**
>
> Autosave does not increment posts.version and does not create a revision or public side effect.

DB-TBL-022 - post_url_history

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Retained independently of Post hard delete; later routing/security authority may govern final purge.

Column	PostgreSQL type	NULL?	Default	Physical meaning
url_history_id	uuid	NO	gen_random_uuid()	History identity.
site_id	uuid	NO	-	Owning site.
post_id	uuid	NO	-	Historical Post UUID; deliberately not FK-cascaded to posts.
slug	text	NO	-	Historical site-scoped slug only; PDF 09 builds final URL.
became_public_at	timestamptz	NO	-	Start of public-addressable episode.
ceased_public_at	timestamptz	YES	NULL	End of episode when known.
created_by_admin_id	uuid	YES	NULL	Actor for manual content change; NULL for system transition.
created_at	timestamptz	NO	now()	History record creation.

Constraints

- PK(url_history_id)
- UQ(site_id,url_history_id)

- FK site_id -> sites
- FK created_by_admin_id -> admin_accounts
- CK ceased_public_at >= became_public_at when non-null

Indexes

- ix_post_url_history__site_post
- ix_post_url_history__site_slug
- ix_post_url_history__site_open_episode

> **IMPLEMENTATION RULE**
>
> No FK to posts is deliberate so routing history can survive terminal Post deletion. Current-slug reuse/redirect precedence remains PDF 09/16 authority.

12. Navigation and Conditional Homepage Persistence

DB-TBL-024 - navigation_menus

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	System/site bootstrap creates HEADER and FOOTER rows; normal Admin does not create arbitrary menu types.

Column	PostgreSQL type	NULL?	Default	Physical meaning
menu_id	uuid	NO	gen_random_uuid()	Menu identity.
site_id	uuid	NO	-	Owning site.
menu_key	text	NO	-	HEADER | FOOTER.
created_at	timestamptz	NO	now()	Bootstrap time.
updated_at	timestamptz	NO	now()	Latest menu metadata mutation.

Constraints

- PK(menu_id)
- UQ(site_id,menu_id)
- FK site_id -> sites
- UQ(site_id,menu_key)
- CK menu_key HEADER/FOOTER

Indexes

- uq_navigation_menus__site_key

> **IMPLEMENTATION RULE**
> Two rows per site are seeded/provisioned; navigation.manage operates on items, not arbitrary new menu classes.

DB-TBL-025 - navigation_items

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Hard delete is normal item removal; audit preserves change history.

Column	PostgreSQL type	NULL?	Default	Physical meaning
item_id	uuid	NO	gen_random_uuid()	Navigation item identity.
site_id	uuid	NO	-	Owning site.
menu_id	uuid	NO	-	Owning Header/Footer menu.
parent_item_id	uuid	YES	NULL	Optional same-menu parent.
label	text	NO	-	Rendered navigation label.
sort_order	integer	NO	0	Sibling order.
is_enabled	boolean	NO	true	Visibility flag.
destination_kind	text	NO	-	HOME | POST | CATEGORY | TAG | AUTHOR | SEARCH | STATIC | EXTERNAL.
target_post_id	uuid	YES	NULL	Same-site Post target.
target_category_id	uuid	YES	NULL	Same-site Category target.

Column	PostgreSQL type	NULL?	Default	Physical meaning
target_tag_id	uuid	YES	NULL	Same-site Tag target.
target_author_profile_id	uuid	YES	NULL	Same-site Author target.
static_page_key	text	YES	NULL	ABOUT | CONTACT | PRIVACY | TERMS | DISCLAIMER.
external_url	text	YES	NULL	Validated external destination input.
created_at	timestamptz	NO	now()	Creation time.
updated_at	timestamptz	NO	now()	Latest mutation.

Constraints

- PK(item_id)
- UQ(site_id,item_id)
- UQ(site_id,menu_id,item_id)
- Composite FK(site_id,menu_id) -> navigation_menus
- Composite self-FK(site_id,menu_id,parent_item_id) -> navigation_items
- Composite FKs for all site-owned targets
- CK sort_order>=0
- CK destination_kind allowed
- CK exactly the destination columns required by destination_kind are populated
- CK static_page_key one of five DOC-05 types when STATIC

Indexes

- ix_navigation_items__site_menu_parent_order
- ix_navigation_items__site_enabled

> **IMPLEMENTATION RULE**
>
> Exact URL generation and supported nesting depth remain PDF 09/14; DB enforces site ownership and fixed public destination classes only.

DB-TBL-026 - homepage_sections

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Release/frontend-contract seeded; normal Admin may update exposed fields but may not create arbitrary sections.

Column	PostgreSQL type	NULL?	Default	Physical meaning
section_id	uuid	NO	gen_random_uuid()	Section registry identity.
site_id	uuid	NO	-	Owning site.
section_key	text	NO	-	Frontend-contract stable key.
section_kind	text	NO	-	POST_SELECTION | CATEGORY_SELECTION | TOGGLE_ONLY.
contract_version	text	NO	-	Frontend contract version that exposed the row.
position_key	text	NO	-	Developer-owned position identifier; not Admin page-builder order.
is_enabled	boolean	NO	true	Conditional section visibility state.
display_label	text	YES	NULL	Approved editable label when contract allows.
max_items	smallint	YES	NULL	Selection limit when applicable.
allow_visibility_edit	boolean	NO	false	Frontend contract flag.

Column	PostgreSQL type	NULL?	Default	Physical meaning
allow_label_edit	boolean	NO	false	Frontend contract flag.
allow_post_selection	boolean	NO	false	Frontend contract flag.
allow_category_selection	boolean	NO	false	Frontend contract flag.
created_at	timestamptz	NO	now()	Contract seed time.
updated_at	timestamptz	NO	now()	Latest permitted mutation.

Constraints

- PK(section_id)
- UQ(site_id,section_id)
- FK site_id -> sites
- UQ(site_id,section_key)
- CK section_kind allowed
- CK max_items positive when present

Indexes

- uq_homepage_sections__site_key
- ix_homepage_sections__site_enabled

> **IMPLEMENTATION RULE**
>
> Baseline seed contains no rows until a site frontend contract exposes conditional FC-NAV-07..10 controls. This table is not a page-builder registry.

DB-TBL-027 - homepage_section_posts

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Selection rows may be inserted/removed only for an exposed Post-selection section.

Column	PostgreSQL type	NULL?	Default	Physical meaning
site_id	uuid	NO	-	Tenant identity.
section_id	uuid	NO	-	Owning homepage section.
post_id	uuid	NO	-	Same-site selected Post.
position	integer	NO	-	Selection order >=0.
created_at	timestamptz	NO	now()	Selection time.

Constraints

- PK(site_id,section_id,post_id)
- UQ(site_id,section_id,position)
- Composite FK(site_id,section_id) -> homepage_sections ON DELETE CASCADE
- Composite FK(site_id,post_id) -> posts
- CK position>=0

Indexes

- ix_homepage_section_posts__site_post

> **IMPLEMENTATION RULE**
>
> Application validates the section contract flags and selection limit; public eligibility is resolved when rendering.

DB-TBL-028 - homepage_section_categories

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Selection rows may be inserted/removed only for an exposed Category-selection section.

Column	PostgreSQL type	NULL?	Default	Physical meaning
site_id	uuid	NO	-	Tenant identity.
section_id	uuid	NO	-	Owning homepage section.
category_id	uuid	NO	-	Same-site selected Category.
position	integer	NO	-	Selection order >=0.
created_at	timestamptz	NO	now()	Selection time.

Constraints

- PK(site_id,section_id,category_id)
- UQ(site_id,section_id,position)
- Composite FK(site_id,section_id) -> homepage_sections ON DELETE CASCADE
- Composite FK(site_id,category_id) -> categories
- CK position>=0

Indexes

- ix_homepage_section_categories__site_category

> **IMPLEMENTATION RULE**
>
> No cross-site feed selection is supported.

13. Site and Platform Settings Persistence

DB-TBL-029 - site_settings

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	One row per site; deleted only with controlled site purge.

Column	PostgreSQL type	NULL?	Default	Physical meaning
site_id	uuid	NO	-	PK and owning site.
logo_media_id	uuid	YES	NULL	Approved site logo asset reference.
favicon_media_id	uuid	YES	NULL	Approved favicon asset reference.
default_social_image_media_id	uuid	YES	NULL	Approved default social image reference.
contact_info	jsonb	NO	'{}'::jsonb	Approved public contact object; no submission workflow.
social_profiles	jsonb	NO	'{}'::jsonb	Validated public social-link map.
copyright_text	text	YES	NULL	Approved footer copyright text.
seo_defaults	jsonb	NO	'{}'::jsonb	Site-wide SEO default inputs; PDF 16 owns semantics/fallback.
analytics_verification	jsonb	NO	'{}'::jsonb	Approved public analytics/verification identifiers only; no secrets.
created_at	timestamptz	NO	now()	Provisioning time.
updated_at	timestamptz	NO	now()	Latest settings mutation.
updated_by_admin_id	uuid	YES	NULL	Latest Admin editor; NULL for provisioning/system.

Constraints

- PK(site_id)
- FK site_id -> sites ON DELETE CASCADE
- Composite FKs for all media refs -> media
- FK updated_by_admin_id -> admin_accounts
- CK JSON fields are objects

Indexes

- ix_site_settings__updated_at

> **IMPLEMENTATION RULE**
>
> Site identity name, locale and timezone are canonical columns on sites and are not duplicated here. Low-level visual theme/code tokens are not stored.

DB-TBL-031 - platform_settings

Property	Locked value
Scope	GLOBAL
RLS	NO
Retention/delete	Rows are migration/seed-defined; runtime Admin may update approved rows but not invent keys.

Column	PostgreSQL type	NULL?	Default	Physical meaning
setting_key	text	NO	-	Stable approved key; primary identifier.
value_jsonb	jsonb	NO	-	Current approved business/platform value.
value_type	text	NO	-	STRING | NUMBER | BOOLEAN | OBJECT | ARRAY.
is_editable	boolean	NO	true	Whether Central Admin may mutate this row.
description	text	NO	-	Human help text.
updated_at	timestamptz	NO	now()	Latest mutation.
updated_by_admin_id	uuid	YES	NULL	Latest Admin editor; NULL for migration/seed.

Constraints

- PK(setting_key)
- FK updated_by_admin_id -> admin_accounts
- CK value_type allowed
- CK JSON type matches value_type where practical

Indexes

- ix_platform_settings__editable

> **IMPLEMENTATION RULE**
>
> Baseline seed intentionally contains no business keys because DOC-01..07 have not frozen any exact global setting. Runtime INSERT/DELETE are not granted; later authorities add approved rows by migration.

14. Redirect Persistence

DB-TBL-030 - redirects

Property	Locked value
Scope	SITE
RLS	YES
Retention/delete	Hard delete is normal redirect removal; audit preserves action history.

Column	PostgreSQL type	NULL?	Default	Physical meaning
redirect_id	uuid	NO	gen_random_uuid()	Redirect identity.
site_id	uuid	NO	-	Owning site.
source_path	text	NO	-	Site-local source path input; PDF 09 owns normalization/precedence.
destination_kind	text	NO	-	INTERNAL_PATH | EXTERNAL_URL.
destination_value	text	NO	-	Validated destination value.
status_code	smallint	NO	-	Required HTTP 3xx code selected by the later redirect/routing authority; DOC-08 persists it but does not choose a default.
is_active	boolean	NO	true	Rule enabled flag.
created_by_admin_id	uuid	NO	-	Creating Admin.
updated_by_admin_id	uuid	YES	NULL	Latest editor.
created_at	timestamptz	NO	now()	Creation time.
updated_at	timestamptz	NO	now()	Latest mutation.

Constraints

- PK(redirect_id)
- UQ(site_id,redirect_id)
- FK site_id -> sites
- UQ(site_id,source_path)
- FK actor IDs -> admin_accounts
- CK destination_kind allowed
- CK status_code BETWEEN 300 AND 399
- CK source_path begins /

Indexes

- uq_redirects__site_source
- ix_redirects__site_active

> **IMPLEMENTATION RULE**
>
> Loop/conflict/canonical interaction remains PDF 09/16; this table persists site ownership and one active source namespace per site.

15. Audit and Site-Deletion-Request Persistence

DB-TBL-032 - audit_events

Property	Locked value
Scope	GLOBAL
RLS	NO
Retention/delete	Append-only; no UPDATE/DELETE grant to runtime roles. Retention period belongs security/operations authority.

Column	PostgreSQL type	NULL?	Default	Physical meaning
audit_id	uuid	NO	gen_random_uuid()	Stable audit event identity.
occurred_at	timestamptz	NO	now()	Event time.
scope	text	NO	-	SITE | GLOBAL.
site_id	uuid	YES	NULL	Required for SITE-scoped event.
actor_type	text	NO	-	ADMIN | SYSTEM.
actor_admin_id	uuid	YES	NULL	Admin identity when actor_type=ADMIN.
event_code	text	NO	-	Stable event code defined by later audit/security authority.
module_key	text	YES	NULL	Operational module grouping.
action_key	text	YES	NULL	Action/capability context.
target_type	text	YES	NULL	Target entity class.
target_id	uuid	YES	NULL	Target stable UUID when applicable.
request_id	text	YES	NULL	Request/correlation identifier.
outcome	text	NO	-	SUCCESS | DENIED | FAILED.
details	jsonb	NO	'{}'::jsonb	Privacy-minimized event details.

Constraints

- PK(audit_id)
- FK site_id -> sites
- FK actor_admin_id -> admin_accounts
- CK scope SITE/GLOBAL
- CK SITE requires site_id; GLOBAL permits null
- CK actor type and actor_admin_id presence
- CK outcome allowed
- CK details object

Indexes

- ix_audit_events__occurred_at
- ix_audit_events__site_time
- ix_audit_events__actor_time
- ix_audit_events__target
- ix_audit_events__event_code_time

> **IMPLEMENTATION RULE**
>
> FC-CAL-08 is not a SITE audit-log permission; Post revision/status history stays in post_revisions. This table backs the GLOBAL Audit module.

DB-TBL-033 - site_deletion_requests

Property	Locked value
Scope	GLOBAL

Property	Locked value
RLS	NO
Retention/delete	Retained as controlled request evidence; final site purge mechanics are later security/deployment authority.

Column	PostgreSQL type	NULL?	Default	Physical meaning
request_id	uuid	NO	gen_random_uuid()	Deletion request identity.
site_id	uuid	NO	-	Target site.
requested_by_admin_id	uuid	NO	-	Authorized requesting Admin.
reason	text	NO	-	Required human reason.
requested_at	timestamptz	NO	now()	Request creation time.
closed_at	timestamptz	YES	NULL	Set when later governed process closes request.
closed_by_admin_id	uuid	YES	NULL	Closing Admin/system representative when applicable.
resolution_note	text	YES	NULL	Controlled closure note; exact workflow states are not invented here.

Constraints

- PK(request_id)
- FK site_id -> sites ON DELETE RESTRICT
- FK actor IDs -> admin_accounts
- CK reason nonblank
- CK closed_by/closed_at pair consistency

Indexes

- ix_site_deletion_requests__site_time
- ix_site_deletion_requests__open

> **IMPLEMENTATION RULE**
>
> The canonical site lifecycle remains sites.status=DELETION_REQUESTED. This table records the request without inventing approval states owned by PDFs 19/24.

11.1 Canonical revision snapshot payload

post_revisions.snapshot is an immutable JSONB object. It records enough content and workflow metadata to understand the resulting durable state while keeping the Post identity and normalized relationship source tables authoritative for current state. The minimum payload is:

```
{
  "content_schema_version": 1,
  "title": "...",
  "excerpt": null,
  "body": {"blocks": [...]},
  "relationships": {
    "author_site_profile_id": "uuid|null",
    "category_ids": ["uuid"],
    "tag_ids": ["uuid"],
    "featured_media_id": "uuid|null",
    "manual_related_post_ids": ["uuid"]
  },
  "seo": {
    "slug": "...", "seo_title": null, "meta_description": null,
    "canonical_override": null, "index_policy": "INHERIT",
    "link_robots_policy": "INHERIT", "og_title": null,
    "og_description": null, "og_image_media_id": null,
    "structured_data_inputs": {}, "breadcrumb_label": null
  },
  "workflow": {
    "status": "DRAFT|SCHEDULED|PUBLISHED|ARCHIVED|TRASH",
    "scheduled_for_utc": null, "schedule_local_input": null,
    "schedule_timezone_id": null, "first_published_at": null,
    "published_at": null, "archived_at": null, "trashed_at": null
  }
```

```
}
```

REVISION RESTORE BOUNDARY

The workflow object is historical context and is not blindly restored. DOC-07 WF-019 restores only authorized editorial/relationship/SEO fields, preserves the current lifecycle status, active schedule and publication timestamps, re-applies all field-level permissions and current validation, then inserts a new revision.

16. Same-Site Relational Integrity

Composite foreign keys are the database-level expression of DOC-04 ISO-INV-08. For every relationship between two site-owned records, the child carries site_id and references a parent UNIQUE(site_id, id) key. A valid UUID from another site is therefore invalid in the current relationship even before application authorization is considered.

Relationship	Database enforcement	Frozen result
posts.author_site_profile_id	posts(site_id,author_site_profile_id) -> author_site_profiles(site_id,author_profile_id)	Author attribution same-site.
posts.featured_media_id / og_image_media_id	posts(site_id,media_id) -> media(site_id,media_id)	Post media same-site.
categories.parent_category_id	categories(site_id,parent) -> categories(site_id,category_id)	Hierarchy cannot cross site.
tags.merged_into_tag_id	tags(site_id,target) -> tags(site_id,tag_id)	Merge target same-site.
post_categories	site_id+post_id and site_id+category_id composite FKs	Association same-site.
post_tags	site_id+post_id and site_id+tag_id composite FKs	Association same-site.
post_related_posts	source and related both composite FKs to posts	Related content same-site.
media_variants	site_id+media_id -> media	Derived media same-site.
author_site_profiles.photo_media_id	site_id+photo_media_id -> media	Author image same-site.
navigation_items targets	site_id + each target id -> owning table	Navigation cannot attach another site object.
homepage selections	site_id+section + site_id+selected object	Conditional homepage selection same-site.
site_settings brand media	site_id+media_id -> media	Brand assets same-site.

16.1 Site ID immutability

Ordinary UPDATE does not move any site-owned row to another site. The runtime repository never exposes site_id as an editable ownership field. RLS WITH CHECK plus the shared reject_site_id_change trigger block a cross-site move even if a malformed payload reaches persistence. A future controlled migration may move data only outside ordinary application credentials and only under a separately approved change.

17. Row-Level Security Architecture

17.1 Canonical policy pattern

```
ALTER TABLE posts ENABLE ROW LEVEL SECURITY;
ALTER TABLE posts FORCE ROW LEVEL SECURITY;

CREATE POLICY rls_posts__site ON posts
  FOR ALL
  USING (site_id = app.current_site_id())
  WITH CHECK (site_id = app.current_site_id());
```

The same semantic policy is applied to all 20 SITE-owned tables. Tables that are immutable/append-only receive narrower GRANTs even though RLS remains active. Public eligibility, Admin permission, object state and business validation remain application concerns layered above tenant RLS.

Table	USING / WITH CHECK	FORCE RLS	Notes
categories	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
tags	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
media	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
media_variants	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.

Table	USING / WITH CHECK	FORCE RLS	Notes
author_site_profiles	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
posts	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
post_categories	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
post_tags	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
post_related_posts	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
post_revisions	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
post_autosaves	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
post_url_history	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
media_usages	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
navigation_menus	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
navigation_items	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
homepage_sections	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
homepage_section_posts	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
homepage_section_categories	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
site_settings	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.
redirects	site_id = app.current_site_id()	YES	Same policy on SELECT/INSERT/UPDATE/DELETE subject to table grants.

17.2 Transaction-local tenant setting

1. Resolve trusted SiteContext before database work.
2. Begin transaction on site-runtime connection.
3. Execute SET LOCAL app.site_id = '<trusted uuid>'.
4. Execute repository statements; RLS and composite FKs enforce tenant visibility/write ownership.
5. Commit or roll back; SET LOCAL disappears automatically.
6. Return connection to pool with no persistent tenant authority.

POOL LEAKAGE RULE

The application must never use persistent SET app.site_id on pooled connections. A connection checkout is not trusted merely because it was previously used by the same site. Every site transaction establishes its own SET LOCAL value from the current resolved SiteContext.

18. Database Roles, Grants and Global-Read Boundary

DB role class	Login?	BYPASSRLS?	Permitted purpose
db_migrator	Deployment only	YES / owner-level as required	Apply versioned migrations, RLS/policy DDL and controlled system seeds. Never used by public/Admin/worker runtime.
db_site_runtime	Runtime pool	NO	Site-scoped SELECT/DML on approved site-owned tables under SET LOCAL app.site_id; approved reads of necessary global registries such as site/domain metadata.
db_platform_runtime	Runtime pool	NO	Authorized GLOBAL table operations: sites/domains/Admin identities/roles/assignments/platform settings; no unrestricted site-owned DML.
db_global_read_owner	NOLOGIN function/view owner	May bypass RLS only inside narrowly approved SECURITY DEFINER global-read functions/views	Supports authorized Network Dashboard/global read surfaces without making missing SiteContext global. Runtime cannot SET ROLE to this owner.

Global create handoff is never performed with a cross-site mutation credential. The application selects/authorizes the destination site, then performs the site-owned create through the normal site-runtime transaction with that SiteContext. Network/global read helpers return ownership identifiers and cannot change ownership.

18.1 Runtime grant restrictions

- No runtime role owns the site-owned tables.
- db_site_runtime receives no CREATE/ALTER/DROP/TRUNCATE privileges.
- permissions and permission_prerequisites are SELECT-only to runtime; mutations are migrations/seeds.
- admin_accounts has no routine DELETE grant.
- post_revisions has no runtime UPDATE/DELETE grant.
- audit_events has INSERT/SELECT as needed but no UPDATE/DELETE grant.
- homepage_sections has SELECT/UPDATE on approved fields through service but no normal INSERT/DELETE; frontend-contract provisioning uses migrations/controlled seed.
- platform_settings permits SELECT/UPDATE of existing approved rows; normal runtime has no INSERT/DELETE.

19. Optimistic Concurrency and Transactional Mutation Rules

19.1 Compare-and-swap

```
UPDATE posts
SET title = :title, body = :body, version = version + 1,
    updated_at = now(), updated_by_admin_id = :actor
WHERE site_id = app.current_site_id()
  AND post_id = :post_id
  AND version = :expected_version
RETURNING version;
```

Exactly one row means the durable mutation owns the current version. Zero rows is treated as conflict/not-found according to API non-disclosure rules; the service must distinguish only where authorized. The service then inserts the formal post_revisions row in the same transaction. No last-write-wins fallback exists.

19.2 Durable mutation transaction

7. Authorize operation and field-level permissions under DOC-02.
8. Resolve SiteContext and begin DB transaction; SET LOCAL app.site_id.
9. Reload/lock or compare current Post version/state as required.
10. Validate same-site relationships and ArticleBody schema.
11. Apply normalized relation changes and media-usage extraction.
12. CAS-update posts and increment version exactly once.
13. Insert one immutable post_revisions snapshot for revision-producing mutation.
14. Insert/update post_url_history when the already-public locator changes as defined by DOC-07; final route handling remains PDF 09.
15. Commit. Only after commit may downstream cache/search/sitemap/public refresh processing be dispatched according to later authorities.

19.3 Permanent Delete

TERMINAL DELETE RULE

Permanent Delete is accepted only after DOC-07 TRASH/state/version/authorization/retention gates succeed. It does not create a new Post revision. The transaction hard-deletes posts and cascade-owned current relations/revisions/autosaves/media-usage rows. post_url_history is deliberately retained independently. Audit accountability remains in audit_events and later PDF 19 rules.

20. Indexing and Query-Shape Contract

Indexes are part of the persistence contract because tenant scoping must remain performant at the expected 100,000+ Post scale. Exact query tuning may add indexes later, but required baseline indexes below must exist unless an approved measured replacement is documented.

Table	Required baseline index / uniqueness object
sites	uq_sites__site_key
sites	ix_sites__status
domains	uq_domains__hostname_normalized
domains	uq_domains__one_active_canonical_per_site
domains	ix_domains__site_id_active
admin_accounts	uq_admin_accounts__email_normalized
admin_accounts	uq_admin_accounts__single_super_admin
admin_accounts	ix_admin_accounts__status
permissions	ix_permissions__scope_module
permission_prerequisites	ix_permission_prerequisites__required
roles	uq_roles__scope_name
roles	ix_roles__scope_status
role_permissions	ix_role_permissions__permission_key
account_site_assignments	ix_account_site_assignments__site_id
account_site_role_assignments	ix_account_site_role_assignments__admin_site
account_site_role_assignments	ix_account_site_role_assignments__role
account_global_role_assignments	ix_account_global_role_assignments__admin
account_global_role_assignments	ix_account_global_role_assignments__role
categories	uq_categories__site_slug
categories	uq_categories__site_name
categories	ix_categories__site_parent
categories	ix_categories__site_status
tags	uq_tags__site_slug
tags	uq_tags__site_name
tags	ix_tags__site_status
media	uq_media__site_object_key
media	ix_media__site_status_created
media	ix_media__site_checksum
media_variants	uq_media_variants__media_key
media_variants	uq_media_variants__site_object_key
author_site_profiles	uq_author_profiles__site_slug
author_site_profiles	ix_author_profiles__site_status
author_site_profiles	ix_author_profiles__site_name

Table	Required baseline index / uniqueness object
posts	uq_posts__site_slug
posts	ix_posts__site_status_published
posts	ix_posts__site_status_scheduled
posts	ix_posts__site_author
posts	ix_posts__site_updated
posts	ix_posts__site_created
post_categories	ix_post_categories__site_category
post_tags	ix_post_tags__site_tag
post_related_posts	ix_post_related_posts__site_related
post_revisions	ix_post_revisions__site_post_created
post_revisions	ix_post_revisions__site_post_revision
post_autosaves	uq_post_autosaves__post_editor
post_autosaves	ix_post_autosaves__editor_saved
post_url_history	ix_post_url_history__site_post
post_url_history	ix_post_url_history__site_slug
post_url_history	ix_post_url_history__site_open_episode
media_usages	ix_media_usages__site_media
media_usages	ix_media_usages__site_post
media_usages	ix_media_usages__site_author
navigation_menus	uq_navigation_menus__site_key
navigation_items	ix_navigation_items__site_menu_parent_order
navigation_items	ix_navigation_items__site_enabled
homepage_sections	uq_homepage_sections__site_key
homepage_sections	ix_homepage_sections__site_enabled
homepage_section_posts	ix_homepage_section_posts__site_post
homepage_section_categories	ix_homepage_section_categories__site_category
site_settings	ix_site_settings__updated_at
redirects	uq_redirects__site_source
redirects	ix_redirects__site_active
platform_settings	ix_platform_settings__editable
audit_events	ix_audit_events__occurred_at
audit_events	ix_audit_events__site_time
audit_events	ix_audit_events__actor_time
audit_events	ix_audit_events__target
audit_events	ix_audit_events__event_code_time
site_deletion_requests	ix_site_deletion_requests__site_time
site_deletion_requests	ix_site_deletion_requests__open

20.1 Site-first query principle

High-volume site-owned access paths put site_id at or near the leading index position. The canonical Post list index is conceptually (site_id, status, published_at DESC, updated_at DESC); the slug locator is (site_id, slug). Queries that intentionally operate globally use the explicit global-read path rather than dropping site_id from ordinary repository SQL.

21. Archive, Delete and Retention Behavior

Record class	Persistence behavior	Locked rule
sites	Lifecycle retained	No ordinary DELETE. DISABLED/ARCHIVED/DELETION_REQUESTED remain rows. Final purge only later governed process.
domains	Managed mapping	Aliases may be removed; site purge cascades domain rows.
admin_accounts	Disable retained	No routine hard delete; audit attribution preserved.
roles	Retire retained	No routine hard delete after use; assignments must be resolved before retirement.
permissions	Registry permanent	Migration-owned; no runtime delete.
account/role assignments	Hard remove	Removal is authoritative revocation; audit preserves change history.
categories/tags/authors	Soft retire	No user-facing restore is implied; historical relationships remain. Hard purge is maintenance-only after safe dependency checks.
media	Soft tombstone	Row/object retention allows safe reference/revision handling; exact physical object purge later PDF 17/19.
posts	TRASH then terminal hard delete	TRASH is recoverable; Permanent Delete hard-deletes Post graph after gates and creates no new revision.
post_revisions	Immutable while Post exists	Cascade on Permanent Delete.
post_autosaves	Ephemeral latest per editor	Upserted; may be removed after durable commit; cascade on Post delete.
post_url_history	Retained independently	Survives Post hard delete by design; later routing/security policy may purge under controlled migration/maintenance.
navigation items/redirects	Hard remove	No restore capability; audit preserves operation.
homepage_sections	Contract retained	Created/removed by controlled frontend-contract release, not routine Admin page-building.
site_settings	Site lifetime	One row per site; deleted only with site purge.
platform_settings	Migration-owned key registry	Values may change; keys added/removed only by migration.
audit_events	Append-only	No runtime update/delete. Exact time retention/partitioning is PDF 19/20 authority.
site_deletion_requests	Retained evidence	Final purge/closure behavior is later security/deployment authority.

NO INVENTED RETENTION PERIODS

DOC-08 freezes whether records are hard-deleted, soft-retained or append-only. It does not invent legal/security time periods that DOC-01..07 never specified. Exact audit/media/backup/privacy retention durations are later PDF 19/20/24 decisions and must be implemented without weakening the physical rules above.

22. Migration and Bootstrap Order

Order	Migration	Required result
DB-MIG-001	bootstrap_database	Enable pgcrypto; create app helper schema/functions; create DB technical roles/grant skeleton.
DB-MIG-002	tenancy_registry	Create sites and domains.
DB-MIG-003	admin_identity_authorization	Create admin_accounts, permissions, permission_prerequisites, roles, role_permissions and assignment tables.

Order	Migration	Required result
DB-MIG-004	taxonomy_media_author	Create categories, tags, media, media_variants and author_site_profiles; add same-site media FKs.
DB-MIG-005	post_core	Create posts with five-state/check/concurrency constraints and same-site author/media FKs.
DB-MIG-006	post_relationships	Create post_categories, post_tags and post_related_posts.
DB-MIG-007	post_history	Create post_revisions, post_autosaves and post_url_history.
DB-MIG-008	media_usage	Create media_usages and usage constraints/indexes.
DB-MIG-009	navigation_homepage	Create navigation_menus/items and homepage section/selection tables.
DB-MIG-010	settings_redirects	Create site_settings, platform_settings and redirects.
DB-MIG-011	audit_deletion_request	Create audit_events and site_deletion_requests.
DB-MIG-012	immutability_and_role_integrity	Create/apply site_id immutability triggers; role-scope/retirement enforcement triggers.
DB-MIG-013	rls_policies	Enable + FORCE RLS on all 20 site-owned tables and create policies.
DB-MIG-014	runtime_grants	Apply runtime table/function grants; remove unsafe default PUBLIC privileges.
DB-MIG-015	required_system_seed	Idempotently seed 69 permissions, prerequisite rows, 14 sites/domains using explicit per-site timezone bootstrap inputs, site_settings and HEADER/FOOTER menus.
DB-MIG-016	validation_gate	Run migration assertions: counts, FK/RLS presence, canonical-domain/site seed integrity and valid nonblank IANA timezone for every initial site. Deployment fails on mismatch.

22.1 Migration rules

- Every schema change is version controlled and applied first in test/staging before production.
- No migration edits a production database by hand outside the migration chain.
- Schema and data migrations are transactional where PostgreSQL permits; non-transactional operations require an explicit safe runbook later PDF 24.
- Destructive migration steps are never combined casually with application deploys; compatibility windows/forward-fix rules are PDF 24.
- Seed code is separate from demo/test fixture code and is safe to rerun.
- A migration that changes a frozen table/constraint/RLS rule must update DOC-08 traceability before release.

23. System and Permission Seed Contract

23.1 Seed classes

Seed class	Runs in production?	Contents	Idempotency key
REQUIRED_SYSTEM	YES	69 permission rows, permission prerequisites, required registry/help metadata.	permission_key / prerequisite pair
INITIAL_ESTATE	YES for initial launch	14 site rows, 14 canonical domain rows, site_settings row per site, HEADER/FOOTER menus.	stable site_id/site_key/domain/menu key
BOOTSTRAP_OWNER	YES once, controlled	Create/designate exactly one Super Admin identity using externally supplied normalized email; no default password/secret stored in seed source.	admin email + unique Super Admin flag
DEVELOPMENT	NO	Non-sensitive demo Admin identities/content/media metadata for local development only.	deterministic fixture UUIDs
ISOLATION_TEST	NO	Deliberately colliding Site-01/Site-02 slugs/names/filenames to prove isolation.	deterministic fixture UUIDs

23.2 Permission registry seed invariant

REGISTRY LOCK

The required seed must produce exactly 69 permission keys: 44 SITE and 25 GLOBAL. It must not create tags.merge, authors.reassign, roles.duplicate, wildcard keys, DENY rows or hidden permissions. Capability composites remain authorization logic from DOC-02, not new registry keys.

23.3 Prerequisite seed behavior

permission_prerequisites materializes the role-bundle prerequisite rules from DOC-02 Section 11. It does not attempt to store the full 161 capability-to-authorization matrix. The Admin role editor/service must validate that every selected permission key has all seeded prerequisites in the same role; it rejects invalid bundles rather than auto-adding grants.

24. Initial 14-Site Seed Contract

The initial estate uses stable deterministic UUIDs so migrations, fixtures, logs and cross-system configuration may refer to the same site identities without deriving identity from mutable domains. These UUIDs are not secrets.

Site key	Stable site_id	Display name	Canonical domain	Status	Timezone
SITE-01	10000000-0000-4000-8000-000000000001	WhatsYourRashi	WhatsYourRashi.in	ACTIVE	REQUIRED CONFIG INPUT
SITE-02	10000000-0000-4000-8000-000000000002	ApkiRashi	ApkiRashi.com	ACTIVE	REQUIRED CONFIG INPUT
SITE-03	10000000-0000-4000-8000-000000000003	DailyRashi360	DailyRashi360.com	ACTIVE	REQUIRED CONFIG INPUT
SITE-04	10000000-0000-4000-8000-000000000004	ZodiacDaily	ZodiacDaily.in	ACTIVE	REQUIRED CONFIG INPUT
SITE-05	10000000-0000-4000-8000-000000000005	DailyHoroscope360	DailyHoroscope360.com	ACTIVE	REQUIRED CONFIG INPUT
SITE-06	10000000-0000-4000-8000-000000000006	MyHoroscope360	MyHoroscope360.com	ACTIVE	REQUIRED CONFIG INPUT
SITE-07	10000000-0000-4000-8000-000000000007	AstroToday360	AstroToday360.com	ACTIVE	REQUIRED CONFIG INPUT
SITE-08	10000000-0000-4000-8000-000000000008	ApkiKundli	ApkiKundli.com	ACTIVE	REQUIRED CONFIG INPUT
SITE-09	10000000-0000-4000-8000-000000000009	MatchYourKundli	MatchYourKundli.com	ACTIVE	REQUIRED CONFIG INPUT
SITE-10	10000000-0000-4000-8000-000000000010	ApkaNakshatra	ApkaNakshatra.com	ACTIVE	REQUIRED CONFIG INPUT
SITE-11	10000000-0000-4000-8000-000000000011	JaniyePanchang	JaniyePanchang.com	ACTIVE	REQUIRED CONFIG INPUT
SITE-12	10000000-0000-4000-8000-000000000012	RashiRemedies	RashiRemedies.com	ACTIVE	REQUIRED CONFIG INPUT
SITE-13	10000000-0000-4000-8000-000000000013	ApkaPandit	ApkaPandit.in	ACTIVE	REQUIRED CONFIG INPUT
SITE-14	10000000-0000-4000-8000-000000000014	VastuShastra360	VastuShastra360.in	ACTIVE	REQUIRED CONFIG INPUT

24.1 Per-site provisioning rows

- Insert one site_settings row with empty approved JSON objects and NULL optional brand assets.
- Insert exactly one navigation_menus row with menu_key=HEADER and one with menu_key=FOOTER.
- Do not seed Posts, Categories, Tags, Authors, Media, roles or homepage sections into production merely to make the UI look populated.
- locale_code/default_language_code remain NULL until product/operator configuration sets them; this is allowed by DOC-04 SiteContext as configured.
- Canonical domain row is globally unique and marked is_canonical=true, is_active=true.

TIMEZONE SEED CONTRACT

BLOG-DOC-01 through BLOG-DOC-07 require each site to have a configured timezone for scheduling/calendar behavior, but they do not freeze the initial timezone values. DOC-08 therefore does not invent them. The 14-site bootstrap must receive one explicit valid IANA timezone identifier per site, and the validation gate fails if any value is missing or invalid.

25. Development and Isolation-Test Seed Contract

25.1 Development fixture boundary

Development fixtures never run in production. They may create synthetic Admin identities and editorial records but contain no real credentials/secrets. Credential setup remains PDF 19/test-harness responsibility.

25.2 Required isolation collision fixture

Site	Post slug	Category	Tag	Author	Original filename	Object key
SITE-01	future-of-ai	Technology	AI	Test Author	hero.jpg	sites/SITE-01/test/hero.jpg
SITE-02	future-of-ai	Technology	AI	Test Author	hero.jpg	sites/SITE-02/test/hero.jpg

The same human values intentionally exist in both sites. The test suite then attempts wrong-site associations and direct RLS reads. Successful coexistence plus failed cross-site access proves that isolation does not depend on globally unique content values.

25.3 Environment guard

```
if ENVIRONMENT == 'production':
    refuse(development_seed)
    refuse(isolation_test_seed)

required_system_seed()       # allowed / idempotent
initial_estate_seed()        # allowed for initial launch
bootstrap_super_admin(input) # explicit controlled input only
```

26. Database Acceptance Criteria and Tests

The database baseline is not accepted until all 88 tests below pass on a clean migrated test database. These tests complement, but do not replace, the broader release/security/isolation QA owned later by PDF 21.

Test ID	Group	Release-blocking assertion
DB-ACC-001	Bootstrap & seed	Clean PostgreSQL instance can apply all migrations in order with no manual DDL.
DB-ACC-002	Bootstrap & seed	Only pgcrypto is required by this baseline; no unapproved extension is assumed.
DB-ACC-003	Bootstrap & seed	All 33 canonical tables exist with the exact names and scope classes in this document.
DB-ACC-004	Bootstrap & seed	Exactly 20 site-owned tables have RLS enabled and forced for runtime roles.
DB-ACC-005	Bootstrap & seed	Exactly 69 permission rows seed successfully: 44 SITE and 25 GLOBAL.
DB-ACC-006	Bootstrap & seed	Permission seed contains no wildcard or deny key.
DB-ACC-007	Bootstrap & seed	All DOC-02 prerequisite pairs seed without unknown permission keys.
DB-ACC-008	Bootstrap & seed	Initial seed creates exactly SITE-01 through SITE-14 with the frozen canonical domains.
DB-ACC-009	Bootstrap & seed	Initial 14-site seed requires a nonblank valid IANA timezone input for every site; DOC-08 does not invent a portfolio-wide timezone value.
DB-ACC-010	Bootstrap & seed	Site bootstrap creates one site_settings row plus HEADER and FOOTER navigation menus for each initial site.
DB-ACC-011	Bootstrap & seed	Production bootstrap contains no default password, shared team login or hidden bypass credential.
DB-ACC-012	Bootstrap & seed	No operational role template is seeded; Super Admin is a designated account flag, not an editable role.
DB-ACC-013	Tenant & RLS	With app.site_id=SITE-01, SELECT on every site-owned table returns only SITE-01 rows.

Test ID	Group	Release-blocking assertion
DB-ACC-014	Tenant & RLS	With no app.site_id, site-owned SELECT/INSERT/UPDATE/DELETE returns/affects no rows under runtime site role.
DB-ACC-015	Tenant & RLS	With app.site_id=SITE-01, INSERT carrying SITE-02 site_id is rejected by RLS WITH CHECK.
DB-ACC-016	Tenant & RLS	Attempt to UPDATE site_id of any site-owned row is rejected by the immutability trigger/RLS path.
DB-ACC-017	Tenant & RLS	A pooled connection that finishes SITE-01 transaction cannot read SITE-01 after starting SITE-02 transaction.
DB-ACC-018	Tenant & RLS	Runtime site credential has no BYPASSRLS, superuser or table-owner bypass capability.
DB-ACC-019	Tenant & RLS	SITE-01 and SITE-02 may store identical post/category/tag/author slugs without uniqueness collision.
DB-ACC-020	Tenant & RLS	Duplicate slug inside the same site is rejected for Posts, Categories, Tags and Author profiles.
DB-ACC-021	Tenant & RLS	Cross-site Post->Category association is rejected by composite FK.
DB-ACC-022	Tenant & RLS	Cross-site Post->Tag association is rejected by composite FK.
DB-ACC-023	Tenant & RLS	Cross-site Post->Author association is rejected by composite FK.
DB-ACC-024	Tenant & RLS	Cross-site Post->featured_media/OG image association is rejected by composite FK.
DB-ACC-025	Tenant & RLS	Cross-site Post->related Post association is rejected by composite FK.
DB-ACC-026	Tenant & RLS	Cross-site Author->photo Media association is rejected by composite FK.
DB-ACC-027	Tenant & RLS	Cross-site Homepage section selection is rejected by composite FK.
DB-ACC-028	Tenant & RLS	Cross-site Navigation target IDs are rejected by composite FK.
DB-ACC-029	Tenant & RLS	Wrong-site URL-history row cannot be read through a SiteContext despite post_id not being FK-bound.
DB-ACC-030	Tenant & RLS	Global tables do not gain site-owned semantics merely because they reference site_id.
DB-ACC-031	Identity & authorization persistence	Only one admin_accounts row can have is_super_admin=true.
DB-ACC-032	Identity & authorization persistence	Case-insensitive duplicate Admin email is rejected.
DB-ACC-033	Identity & authorization persistence	Account lifecycle accepts only INVITED, ACTIVE or DISABLED.
DB-ACC-034	Identity & authorization persistence	SITE role cannot receive GLOBAL permission because role_permissions composite scope FK rejects it.
DB-ACC-035	Identity & authorization persistence	GLOBAL role cannot receive SITE permission for the same reason.
DB-ACC-036	Identity & authorization persistence	SITE role assignment cannot exist without account_site_assignments membership.
DB-ACC-037	Identity & authorization persistence	Deleting an account_site_assignments row cascades its SITE role assignments.
DB-ACC-038	Identity & authorization persistence	GLOBAL role assignment does not create an account_site_assignments row.
DB-ACC-039	Identity & authorization persistence	Nested roles cannot be stored because no role-inheritance table/relationship exists.
DB-ACC-040	Identity & authorization persistence	Explicit deny cannot be stored because no deny column/table exists.
DB-ACC-041	Identity & authorization persistence	Permission registry cannot accept an unknown role permission because FK rejects it.
DB-ACC-042	Identity & authorization persistence	Role retirement is blocked while assignments remain.

Test ID	Group	Release-blocking assertion
DB-ACC-043	Identity & authorization persistence	Role scope mutation is blocked after the role has permissions or assignments.
DB-ACC-044	Post/content persistence	Post status accepts only DRAFT/SCHEDULED/PUBLISHED/ARCHIVED/TRASH.
DB-ACC-045	Post/content persistence	SCHEDULED Post requires scheduled_for_utc, schedule_local_input and schedule_timezone_id together.
DB-ACC-046	Post/content persistence	Non-SCHEDULED Post cannot retain active schedule fields.
DB-ACC-047	Post/content persistence	Post version is positive and compare-and-swap update with stale expected_version affects zero rows.
DB-ACC-048	Post/content persistence	Successful durable mutation increments version exactly once and inserts one matching formal revision.
DB-ACC-049	Post/content persistence	Initial Draft creation inserts revision_number=1 tied to resulting Post version.
DB-ACC-050	Post/content persistence	Revision rows are immutable to runtime application UPDATE/DELETE.
DB-ACC-051	Post/content persistence	Revision numbers and resulting_post_version are unique and monotonic per Post.
DB-ACC-052	Post/content persistence	Autosave upserts one latest row per site/Post/editor and does not change Post version.
DB-ACC-053	Post/content persistence	Successful durable commit may delete that editor autosave without touching other editors autosaves.
DB-ACC-054	Post/content persistence	Revision Restore writes a new Post version and new revision; historical revision rows are unchanged.
DB-ACC-055	Post/content persistence	Permanent Delete from TRASH removes Post and cascade-owned relations/revisions/autosaves but creates no new revision.
DB-ACC-056	Post/content persistence	post_url_history is not cascade-deleted with Post and remains site-owned/RLS-protected.
DB-ACC-057	Post/content persistence	Article body DB check requires an object containing a blocks array.
DB-ACC-058	Post/content persistence	Raw HTML/page-builder tables do not exist; ArticleBody remains structured JSONB only.
DB-ACC-059	Post/content persistence	Body MEDIA usage extraction populates media_usages inside the same durable commit transaction.
DB-ACC-060	Post/content persistence	Retired author cannot be newly assigned by application validation while historical Post FK remains valid.
DB-ACC-061	Post/content persistence	Tag merge can move associations to target without duplicate rows and then retire source tag.
DB-ACC-062	Post/content persistence	Trash does not physically delete a Post; status/TRASH timestamp remain recoverable.
DB-ACC-063	Post/content persistence	Restore from TRASH returns persistence to DRAFT only; database has no hidden published restore state.
DB-ACC-064	Post/content persistence	PUBLISHED Revision Restore persists same status and downstream refresh is triggered by service/outbox layer, not by weakening DB isolation.
DB-ACC-065	Post/content persistence	Bulk mutation applies per-Post version/revision rules; DB relations remain same-site for each target.
DB-ACC-066	Navigation/settings/media/operations	Each initial site has exactly one HEADER and one FOOTER navigation_menus row.
DB-ACC-067	Navigation/settings/media/operations	Navigation STATIC destination accepts only ABOUT/CONTACT/PRIVACY/TERMS/DISCLAIMER keys.
DB-ACC-068	Navigation/settings/media/operations	Navigation item destination check rejects contradictory multiple target columns.
DB-ACC-069	Navigation/settings/media/operations	Homepage section rows cannot be inserted by normal Admin runtime; they originate from release/frontend contract provisioning.
DB-ACC-070	Navigation/settings/media/operations	Baseline homepage_sections seed is empty until a frontend contract exposes conditional controls.

Test ID	Group	Release-blocking assertion
DB-ACC-071	Navigation/settings/media/operations	Site Settings brand-media references must point to same-site media.
DB-ACC-072	Navigation/settings/media/operations	Site Settings JSON objects reject non-object values.
DB-ACC-073	Navigation/settings/media/operations	Platform Settings runtime cannot INSERT arbitrary new setting keys.
DB-ACC-074	Navigation/settings/media/operations	Media object key is unique within site and media variant keys are unique per media.
DB-ACC-075	Navigation/settings/media/operations	Media soft-delete cannot be bypassed into another site namespace.
DB-ACC-076	Navigation/settings/media/operations	Redirect source_path uniqueness is per site, so /old-page may differ across sites.
DB-ACC-077	Navigation/settings/media/operations	Same hostname cannot map to two sites because hostname_normalized is globally unique.
DB-ACC-078	Navigation/settings/media/operations	At most one active canonical domain exists per site.
DB-ACC-079	Navigation/settings/media/operations	Audit events are append-only to runtime roles and SITE-scope events require site_id.
DB-ACC-080	Navigation/settings/media/operations	Site deletion request cannot hard-delete a site; sites table remains lifecycle authority.
DB-ACC-081	Development & isolation fixtures	Development seed cannot execute when environment is marked production.
DB-ACC-082	Development & isolation fixtures	Isolation-test seed creates the same slug future-of-ai in SITE-01 and SITE-02 successfully.
DB-ACC-083	Development & isolation fixtures	Isolation-test seed creates identical category/tag/author names in SITE-01 and SITE-02 successfully.
DB-ACC-084	Development & isolation fixtures	Isolation-test seed uses identical media original filename in both sites while object_key namespaces differ.
DB-ACC-085	Development & isolation fixtures	Isolation-test SQL proves SITE-01 cannot attach SITE-02 category/tag/author/media/Post IDs.
DB-ACC-086	Development & isolation fixtures	Isolation-test SQL proves SITE-01 RLS context cannot read SITE-02 post/revision/autosave/url-history rows.
DB-ACC-087	Development & isolation fixtures	All seed scripts are idempotent by stable key/UUID and do not duplicate permission/site rows.
DB-ACC-088	Development & isolation fixtures	Migration rollback/forward-fix test leaves schema version and seed registry internally consistent.

27. Traceability and Upstream Correlation

DOC-08 does not duplicate the 161-capability catalog or 69-key authorization semantics as independent policy. It persists the entities and constraints required by them. The matrix below records the major upstream-to-database contract links; PDF 22 will later expand them into full requirement-to-implementation traceability.

Req ID	Frozen source	Persistence decision	Physical owner
DB-REQ-001	DOC-01 INV-001/SC-01	Persist exactly 14 initial site identities without hard-coded ceiling	sites + initial site seed
DB-REQ-002	DOC-01 INV-003/INV-017	Centralized PostgreSQL with version-controlled migrations/seeds	entire schema + migration chain
DB-REQ-003	DOC-01 INV-005/018	Missing SiteContext never broadens scope; leakage release-blocking	RLS + app.current_site_id() + composite FKs
DB-REQ-004	DOC-01 SC-06	Same slug/name may repeat across different sites	site-aware unique constraints
DB-REQ-005	DOC-02 registry lock	Persist exactly 69 fixed permission keys	permissions seed
DB-REQ-006	DOC-02 site assignment model	Membership separate from permissions	account_site_assignments + role assignment tables
DB-REQ-007	DOC-02 role scope model	SITE/GLOBAL role scopes cannot mix permission scopes	roles + role_permissions composite FKs
DB-REQ-008	DOC-02 account lifecycle	Persist INVITED/ACTIVE/DISABLED; one Super Admin	admin_accounts

Req ID	Frozen source	Persistence decision	Physical owner
DB-REQ-009	DOC-02 no nested roles/deny/wildcards	No persistence structures for inheritance/deny/wildcard grants	schema absence + exact permission FK
DB-REQ-010	DOC-03 FC-CAT	Persist hierarchical site-owned Categories and assignment integrity	categories + post_categories
DB-REQ-011	DOC-03 FC-TAG	Persist site-owned Tags, merge target and assignment integrity	tags + post_tags
DB-REQ-012	DOC-03 FC-AUTH	Persist site-owned Author profiles and retirement/history	author_site_profiles + posts.author_site_profile_id
DB-REQ-013	DOC-03 FC-MED	Persist media, variants and usage tracking	media + media_variants + media_usages
DB-REQ-014	DOC-03 FC-NAV	Persist Header/Footer menu data and conditional homepage controls	navigation_* + homepage_*
DB-REQ-015	DOC-03 FC-SEO	Persist Post/category/author/site SEO inputs only; algorithms later	posts/categories/authors/site_settings
DB-REQ-016	DOC-03 FC-RED	Persist site-owned redirect inputs	redirects
DB-REQ-017	DOC-03 FC-USR/ROLE/PERM	Persist Admin identities/assignments/roles/fixed registry	auth tables
DB-REQ-018	DOC-03 FC-AUD	Persist immutable global audit history	audit_events
DB-REQ-019	DOC-03 FC-PLAT	Persist only approved global setting rows	platform_settings
DB-REQ-020	DOC-04 ISO-INV-01..13	Physical tenant boundary and RLS	site_id columns + RLS + runtime DB role rules
DB-REQ-021	DOC-04 ISO-INV-08/15	Same-site relationships	composite site_id FKs
DB-REQ-022	DOC-04 ISO-INV-14..18	Derived systems preserve site identity	media/URL/history rows; later search/cache/job schemas constrained
DB-REQ-023	DOC-04 ISO-INV-25	No ordinary cross-site ownership move	site_id immutability trigger
DB-REQ-024	DOC-05 no generic Pages	Do not create generic pages/static_pages table	schema exclusion
DB-REQ-025	DOC-05 five static pages	Allow navigation references by fixed static_page_key only	navigation_items check constraint
DB-REQ-026	DOC-06 19 modules/36 pages	No page/module table needed; DB persists underlying entities only	schema boundary
DB-REQ-027	DOC-07 PST-FLD-001..035	Physical Post mapping	posts + normalized relation tables
DB-REQ-028	DOC-07 12 block types	Persist ArticleBody as versioned JSONB; no block/page-builder tables	posts.body + revision/autosave snapshots
DB-REQ-029	DOC-07 five states	Persist exact status values and lifecycle timestamps	posts checks
DB-REQ-030	DOC-07 revisions/autosave	Separate immutable durable revisions from private autosave	post_revisions + post_autosaves
DB-REQ-031	DOC-07 URL history	Persist site-owned slug history independent of route construction	post_url_history
DB-REQ-032	DOC-07 optimistic concurrency	Persist monotonic version and compare-and-swap pattern	posts.version + update rule
DB-REQ-033	DOC-07 Permanent Delete	Hard-delete Post graph only after TRASH/gates; no delete revision	ON DELETE policy + audit/url-history retention
DB-REQ-034	DOC-07 scheduling	Persist absolute UTC + local input + timezone together	posts schedule fields/check
DB-REQ-035	DOC-07 same-site relations	Physical relationship integrity	composite FKs across Post/taxonomy/author/media/related

28. Downstream Dependencies and Deliberate Exclusions

Decision intentionally not owned here	Later authority	DOC-08 boundary
Exact URL/slug normalization, route history resolution, redirect precedence	PDF 09/16	DOC-08 stores slug/source inputs and site-aware uniqueness/history only.
Admin/Public page behavior/state	PDF 10/11	Page IDs are not database entities.
Public page composition/data projection	PDF 12/13	No response/view-model tables are created.

Decision intentionally not owned here	Later authority	DOC-08 boundary
Frontend integration/theming	PDF 14/26/27	No site UI components/theme token rows are stored here.
HTTP endpoints/idempotency error contracts	PDF 15	No API transport table is invented.
Search ranking/index implementation	PDF 16	No search_documents table is frozen here; any later DB-backed search table must carry site_id/RLS.
Object-storage processing/variant catalog	PDF 17	media/media_variants persist identities; processing rules remain later.
Queue/cache/job payloads	PDF 18	No jobs/cache table is invented; any persisted job later must carry/revalidate site_id.
Password hashes, MFA, reset/invite/session tokens, secret storage	PDF 19	admin_accounts stores identity/lifecycle only.
Audit/privacy retention periods, partition cadence	PDF 19/20	audit_events is append-only; time policy later.
Performance capacity/RPO/RTO/backup topology	PDF 20/24	Required baseline indexes exist, but operational budgets/topology later.
ORM/repository/folder implementation	PDF 23	This PDF is ORM-neutral PostgreSQL authority.
Docker/Coolify migration execution mechanics	PDF 24	Migration order is frozen; deployment orchestration later.

EXPLICIT NON-TABLES

The v1 baseline deliberately has NO generic pages/static_pages table, NO capabilities table, NO role inheritance table, NO deny-permission table, NO cross-site shared taxonomy/media table, NO sessions/secrets table, NO search index table, NO queue/cache table and NO frontend/theme table. Their absence is intentional and prevents accidental scope expansion.

29. Cross-Document Consistency Gate and Final Lock

Upstream	Result	Verification
DOC-01	PASS	One Central Admin/shared backend/centralized PostgreSQL; 14 site IDs; config-driven beyond 14; no generic page builder or duplicated DB stacks.
DOC-02	PASS	Exactly 69 permission keys; three account states; SITE/GLOBAL role separation; site assignment distinct from permissions; one Super Admin; no nested role/deny/wildcard persistence.
DOC-03	PASS	Persistence exists for all frozen data-bearing capability families without adding a module/capability. Conditional homepage persistence is contract-seeded, not a page builder.
DOC-04	PASS	All 20 site-owned tables carry site_id and RLS; same-site composite FKs; no implicit global fallback; normal runtime no BYPASSRLS; URL/media/navigation isolation preserved.
DOC-05	PASS	No generic Pages table. Five static page classes are represented only as fixed navigation target keys when needed.
DOC-06	PASS	No Admin page/module persistence is invented; underlying entities support the frozen 19-module/36-page IA.
DOC-07	PASS	All 35 logical Post fields physically mapped; exact five statuses; 12-block ArticleBody persists as JSONB; revisions/autosaves/URL history/concurrency/Permanent Delete semantics preserved.

CONSISTENCY VERDICT

No genuine contradiction was found between the DOC-08 persistence model and BLOG-DOC-01 through BLOG-DOC-07. DOC-08 therefore does not reopen or modify any frozen upstream document. Any later implementation that requires a different

ownership model, Post status, permission key, page type or cross-site relation must be treated as a controlled specification change.

29.1 Freeze checklist

Freeze ID	Condition
DB-FZ-01	33 canonical tables and scope counts accepted.
DB-FZ-02	69 permission seed / prerequisite seed accepted.
DB-FZ-03	14 deterministic site/domain seed accepted.
DB-FZ-04	All 20 site-owned RLS policies accepted.
DB-FZ-05	Same-site composite FK model accepted.
DB-FZ-06	Post/revision/autosave/URL-history persistence accepted.
DB-FZ-07	Migration/bootstrap order accepted.
DB-FZ-08	Retention/hard-delete/soft-retire boundaries accepted.
DB-FZ-09	Database acceptance suite accepted.
DB-FZ-10	No deliberate exclusion is accidentally implemented as a hidden table/feature.

Appendix A - Canonical 33-Table Registry

Table ID	Table	Scope	RLS	Retention / delete class
DB-TBL-001	sites	GLOBAL	NO	Lifecycle retained; no ordinary hard delete.
DB-TBL-002	domains	GLOBAL	NO	Rows retained while mapped; aliases may be removed under global domain management.
DB-TBL-003	admin_accounts	GLOBAL	NO	Disable instead of routine hard delete; historical identity retained.
DB-TBL-004	permissions	GLOBAL	NO	Migration/seed-owned; runtime Admin is read-only.
DB-TBL-005	permission_prerequisites	GLOBAL	NO	Migration/seed-owned; delete only through controlled permission-registry migration.
DB-TBL-006	roles	GLOBAL	NO	Retired, not routine hard-deleted after use.
DB-TBL-007	role_permissions	GLOBAL	NO	Rows change only through authorized role edits; removed rows are audited externally.
DB-TBL-008	account_site_assignments	GLOBAL	NO	Hard delete represents assignment removal; audit preserves history.
DB-TBL-009	account_site_role_assignments	GLOBAL	NO	Hard delete represents role unassignment; audit preserves history.
DB-TBL-010	account_global_role_assignments	GLOBAL	NO	Hard delete represents role unassignment; audit preserves history.
DB-TBL-011	categories	SITE	YES	Soft-retired in normal product flow; no routine hard purge.
DB-TBL-012	tags	SITE	YES	Soft-retired; tag merge retires the source after relationship rewrite.
DB-TBL-013	media	SITE	YES	Soft-deleted/tombstoned in normal product flow; physical object purge is later media/security policy.
DB-TBL-014	media_variants	SITE	YES	Cascade with media tombstone/purge policy; variants are derived.
DB-TBL-015	author_site_profiles	SITE	YES	Soft-retired; historical Post attribution remains.
DB-TBL-016	posts	SITE	YES	TRASH is soft state; authorized Permanent Delete hard-deletes the Post after gates. URL history may outlive the Post.
DB-TBL-017	post_categories	SITE	YES	Cascade with Post permanent delete; category retirement does not erase historical association automatically.
DB-TBL-018	post_tags	SITE	YES	Cascade with Post permanent delete; tag merge rewrites rows transactionally.
DB-TBL-019	post_related_posts	SITE	YES	Cascade with source Post permanent delete; related target delete is RESTRICT until relation is removed.
DB-TBL-020	post_revisions	SITE	YES	Immutable while Post exists; cascades on authorized Permanent Delete.
DB-TBL-021	post_autosaves	SITE	YES	Ephemeral latest working copy per Post/editor; removed after successful durable commit or Post delete.
DB-TBL-022	post_url_history	SITE	YES	Retained independently of Post hard delete; later routing/security authority may govern final purge.

Table ID	Table	Scope	RLS	Retention / delete class
DB-TBL-023	media_usages	SITE	YES	Derived usage rows removed/rebuilt with owning references; media deletion checks this table.
DB-TBL-024	navigation_menus	SITE	YES	System/site bootstrap creates HEADER and FOOTER rows; normal Admin does not create arbitrary menu types.
DB-TBL-025	navigation_items	SITE	YES	Hard delete is normal item removal; audit preserves change history.
DB-TBL-026	homepage_sections	SITE	YES	Release/frontend-contract seeded; normal Admin may update exposed fields but may not create arbitrary sections.
DB-TBL-027	homepage_section_posts	SITE	YES	Selection rows may be inserted/removed only for an exposed Post-selection section.
DB-TBL-028	homepage_section_categories	SITE	YES	Selection rows may be inserted/removed only for an exposed Category-selection section.
DB-TBL-029	site_settings	SITE	YES	One row per site; deleted only with controlled site purge.
DB-TBL-030	redirects	SITE	YES	Hard delete is normal redirect removal; audit preserves action history.
DB-TBL-031	platform_settings	GLOBAL	NO	Rows are migration/seed-defined; runtime Admin may update approved rows but not invent keys.
DB-TBL-032	audit_events	GLOBAL	NO	Append-only; no UPDATE/DELETE grant to runtime roles. Retention period belongs security/operations authority.
DB-TBL-033	site_deletion_requests	GLOBAL	NO	Retained as controlled request evidence; final site purge mechanics are later security/deployment authority.

Appendix B - 69-Permission Seed Registry

Permission key	Scope	Module	Risk
dashboard.view	SITE	Dashboard	Standard
posts.view	SITE	Posts	Standard
posts.create	SITE	Posts	Standard
posts.edit	SITE	Posts	Standard
posts.preview	SITE	Posts	Sensitive
posts.duplicate	SITE	Posts	Standard
posts.revisions.view	SITE	Posts	Standard
posts.revisions.restore	SITE	Posts	Sensitive
posts.publish	SITE	Posts	Sensitive
posts.schedule	SITE	Posts	Sensitive
posts.archive	SITE	Posts	Sensitive
posts.trash	SITE	Posts	Destructive
posts.restore	SITE	Posts	Sensitive
posts.delete	SITE	Posts	Destructive
categories.view	SITE	Categories	Standard

Permission key	Scope	Module	Risk
categories.create	SITE	Categories	Standard
categories.edit	SITE	Categories	Standard
categories.delete	SITE	Categories	Destructive
tags.view	SITE	Tags	Standard
tags.create	SITE	Tags	Standard
tags.edit	SITE	Tags	Standard
tags.delete	SITE	Tags	Destructive
authors.view	SITE	Authors	Standard
authors.create	SITE	Authors	Standard
authors.edit	SITE	Authors	Standard
authors.retire	SITE	Authors	Sensitive
media.view	SITE	Media	Standard
media.upload	SITE	Media	Standard
media.edit	SITE	Media	Standard
media.replace	SITE	Media	Sensitive
media.delete	SITE	Media	Destructive
navigation.view	SITE	Navigation	Standard
navigation.manage	SITE	Navigation	Sensitive
seo.view	SITE	SEO	Standard
seo.edit	SITE	SEO	Sensitive
redirects.view	SITE	Redirects	Standard
redirects.create	SITE	Redirects	Sensitive
redirects.edit	SITE	Redirects	Sensitive
redirects.delete	SITE	Redirects	Destructive
search.view	SITE	Search	Standard
search.reindex	SITE	Search	Sensitive
calendar.view	SITE	Editorial Calendar	Standard
site_settings.view	SITE	Site Settings	Standard
site_settings.edit	SITE	Site Settings	Sensitive
network_dashboard.view	GLOBAL	Network	Standard
sites.view	GLOBAL	Sites	Standard
sites.create	GLOBAL	Sites	Sensitive
sites.edit	GLOBAL	Sites	Sensitive
sites.enable	GLOBAL	Sites	Sensitive
sites.disable	GLOBAL	Sites	Destructive
sites.archive	GLOBAL	Sites	Destructive
sites.deletion_request	GLOBAL	Sites	Destructive
sites.manage_domains	GLOBAL	Sites	Destructive
admin_users.view	GLOBAL	Admin Users	Standard
admin_users.invite	GLOBAL	Admin Users	Sensitive
admin_users.edit	GLOBAL	Admin Users	Sensitive
admin_users.enable	GLOBAL	Admin Users	Sensitive
admin_users.disable	GLOBAL	Admin Users	Destructive

Permission key	Scope	Module	Risk
admin_users.manage_site_assignments	GLOBAL	Admin Users	Destructive
roles.view	GLOBAL	Roles	Standard
roles.create	GLOBAL	Roles	Sensitive
roles.edit	GLOBAL	Roles	Destructive
roles.retire	GLOBAL	Roles	Destructive
roles.assign	GLOBAL	Roles	Destructive
audit.view	GLOBAL	Audit	Sensitive
audit.export	GLOBAL	Audit	Sensitive
platform_settings.view	GLOBAL	Platform Settings	Sensitive
platform_settings.edit	GLOBAL	Platform Settings	Destructive
permissions.view	GLOBAL	Permissions	Standard

Registry count check: 69 total = 44 SITE + 25 GLOBAL. The seed is authoritative only for persistence of the fixed registry; DOC-02 remains the semantic authorization authority.

Appendix C - Permission-Prerequisite Seed Registry

Permission key	Required prerequisite key
admin_users.disable	admin_users.view
admin_users.edit	admin_users.view
admin_users.enable	admin_users.view
admin_users.invite	admin_users.view
admin_users.manage_site_assignments	admin_users.view
audit.export	audit.view
authors.create	authors.view
authors.edit	authors.view
authors.retire	authors.view
categories.create	categories.view
categories.delete	categories.view
categories.edit	categories.view
media.delete	media.view
media.edit	media.view
media.replace	media.view
media.upload	media.view
navigation.manage	navigation.view
platform_settings.edit	platform_settings.view
posts.archive	posts.edit
posts.archive	posts.view
posts.create	posts.view
posts.delete	posts.view
posts.duplicate	posts.create
posts.duplicate	posts.view
posts.edit	posts.view
posts.publish	posts.edit

Permission key	Required prerequisite key
posts.publish	posts.view
posts.restore	posts.edit
posts.restore	posts.view
posts.revisions.restore	posts.edit
posts.revisions.restore	posts.revisions.view
posts.revisions.restore	posts.view
posts.revisions.view	posts.view
posts.schedule	posts.edit
posts.schedule	posts.view
posts.trash	posts.edit
posts.trash	posts.view
redirects.create	redirects.view
redirects.delete	redirects.view
redirects.edit	redirects.view
roles.assign	roles.view
roles.create	roles.view
roles.edit	roles.view
roles.retire	roles.view
search.reindex	search.view
seo.edit	seo.view
site_settings.edit	site_settings.view
sites.archive	sites.view
sites.create	sites.view
sites.deletion_request	sites.view
sites.disable	sites.view
sites.edit	sites.view
sites.enable	sites.view
sites.manage_domains	sites.view
tags.create	tags.view
tags.delete	tags.view
tags.edit	tags.view

Prerequisite-pair count in this persistence seed: 57. These rows materialize DOC-02 role-bundle prerequisites; they are not a substitute for capability-specific composite authorization.

Appendix D - 14-Site Seed Registry

Site	site_id	Display	Canonical domain
SITE-01	10000000-0000-4000-8000-000000000001	WhatsYourRashi	WhatsYourRashi.in
SITE-02	10000000-0000-4000-8000-000000000002	ApkiRashi	ApkiRashi.com
SITE-03	10000000-0000-4000-8000-000000000003	DailyRashi360	DailyRashi360.com
SITE-04	10000000-0000-4000-8000-000000000004	ZodiacDaily	ZodiacDaily.in

Site	site_id	Display	Canonical domain
SITE-05	10000000-0000-4000-8000-000000000005	DailyHoroscope360	DailyHoroscope360.com
SITE-06	10000000-0000-4000-8000-000000000006	MyHoroscope360	MyHoroscope360.com
SITE-07	10000000-0000-4000-8000-000000000007	AstroToday360	AstroToday360.com
SITE-08	10000000-0000-4000-8000-000000000008	ApkiKundli	ApkiKundli.com
SITE-09	10000000-0000-4000-8000-000000000009	MatchYourKundli	MatchYourKundli.com
SITE-10	10000000-0000-4000-8000-000000000010	ApkaNakshatra	ApkaNakshatra.com
SITE-11	10000000-0000-4000-8000-000000000011	JaniyePanchang	JaniyePanchang.com
SITE-12	10000000-0000-4000-8000-000000000012	RashiRemedies	RashiRemedies.com
SITE-13	10000000-0000-4000-8000-000000000013	ApkaPandit	ApkaPandit.in
SITE-14	10000000-0000-4000-8000-000000000014	VastuShastra360	VastuShastra360.in

Appendix E - RLS Policy Matrix

Table ID	Table	Policy	Site-id immutable?
DB-TBL-011	categories	site_id = app.current_site_id()	YES
DB-TBL-012	tags	site_id = app.current_site_id()	YES
DB-TBL-013	media	site_id = app.current_site_id()	YES
DB-TBL-014	media_variants	site_id = app.current_site_id()	YES
DB-TBL-015	author_site_profiles	site_id = app.current_site_id()	YES
DB-TBL-016	posts	site_id = app.current_site_id()	YES
DB-TBL-017	post_categories	site_id = app.current_site_id()	YES
DB-TBL-018	post_tags	site_id = app.current_site_id()	YES
DB-TBL-019	post_related_posts	site_id = app.current_site_id()	YES
DB-TBL-020	post_revisions	site_id = app.current_site_id()	YES
DB-TBL-021	post_autosaves	site_id = app.current_site_id()	YES
DB-TBL-022	post_url_history	site_id = app.current_site_id()	YES
DB-TBL-023	media_usages	site_id = app.current_site_id()	YES
DB-TBL-024	navigation_menus	site_id = app.current_site_id()	YES
DB-TBL-025	navigation_items	site_id = app.current_site_id()	YES
DB-TBL-026	homepage_sections	site_id = app.current_site_id()	YES
DB-TBL-027	homepage_section_posts	site_id = app.current_site_id()	YES
DB-TBL-028	homepage_section_categories	site_id = app.current_site_id()	YES
DB-TBL-029	site_settings	site_id = app.current_site_id()	YES
DB-TBL-030	redirects	site_id = app.current_site_id()	YES

Appendix F - One-Page Developer Execution Reference

DATABASE SHAPE

33 tables. 20 site-owned tables carry non-null site_id and FORCE RLS. 13 global/system tables are accessed only through explicit platform paths. One DB; never 14 cloned schemas/databases.

Question	Locked answer
Tenant transaction	BEGIN -> SET LOCAL app.site_id=<trusted uuid> -> scoped SQL -> COMMIT/ROLLBACK. Missing setting = no tenant rows.
Same-site relation	Always carry child.site_id and use composite FK to parent(site_id,id).
Post write	Authorize -> validate -> CAS UPDATE version -> normalized relations/media usage -> INSERT revision -> URL history when applicable -> commit.
Autosave	Upsert one row per site/Post/editor; does not change Post.version or public state.
Permanent Delete	TRASH only after higher-layer gates; hard delete Post graph, no new revision, retain URL history/audit.
Role grant	role_permissions scope must match role scope and permission scope; exact key only.
Site membership	account_site_assignments first; SITE role assignment FK depends on it.
Seed	69 permissions + prerequisites + 14 sites/domains + one site_settings + HEADER/FOOTER menus per site. No default role/template/password.
Forbidden	No generic Pages, capability, deny, nested-role, cross-site-share, session/secrets, search-index, queue/cache or frontend/theme tables in this baseline.
Schema change	Migration only. If it changes frozen ownership/field/state/permission semantics, reopen the owning specification before code.

DEVELOPER LOCK

Implementation is complete only when the physical schema, required seeds, RLS/grants, migration chain and database acceptance tests all match this document. An ORM model or migration that omits a listed constraint is not an equivalent implementation simply because application code appears to work.